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Curriculum-Grounded Retrieval-Augmented Generation for Terminale C Mathematics Self-Tutoring in Côte d'Ivoire

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Executive Summary

Context and problem. In Côte d'Ivoire the baccalauréat gates higher education, and its science-intensive série C track is the most selective; yet support is unequal — qualified teachers are scarce, and a competent private tutor costs about a month's minimum wage (≈ 110 € [Gouv.CI, 2023]). Large language models offer an always-available, low-cost tutor, but a general-purpose model applied naïvely to high-stakes curriculum mathematics drifts from the prescribed Ivorian curriculum (methods or notation the student was not taught) and can hallucinate fluent error precisely where a learner is least able to catch it. This thesis asks what form of retrieval-augmented generation (RAG) can be adapted to such a curriculum and — the harder question in a low-resource setting — how the resulting tutor can be evaluated rigorously when domain-expert labels are scarce.

Approach. The work builds and evaluates a RAG tutor grounded in the CIAM Terminale C corpus (the canonical regional textbook), scoped to non-geometry, text-only content: a content-type-tagged corpus of 722 chunks with a validated OCR pipeline and data card; a four-variant retrieval stack (BM25, dense multilingual-E5, their RRF hybrid, and a novel pedagogical retriever enforcing a theory-plus-worked-example mix); and a human-calibrated, model-independent relevance-evaluation framework. Of four hypotheses, three were executed on data in hand; H4 (a grounded frontier-versus-small-model comparison) was only partly completed, awaiting human-expert calibration. H3 was deliberately recast: rather than have an LLM grade complete tutoring responses (needing a (query, response) gold set that could not be collected in time), the LLM acts as a relevance assessor over (query, chunk) pairs — the same object a mixed panel of teachers and Terminale C students already labelled.

Findings. H1 was supported — the hybrid beat dense-only retrieval by +0.049 nDCG@5 (95% paired-bootstrap CI [+0.009, +0.093]), robust across every annotator-perturbation scenario, though the gain is carried by the lexical channel and partly reflects vocabulary overlap between the synthesised evaluation queries and the corpus (a documented threat to validity). The H2 retrieval-side test was not met: the pedagogical retriever trailed the hybrid slightly (−0.034 Recall@5, CI [−0.061, −0.007]) — a small, relevance-neutral cost of enforcing the mix, read as a necessary-condition check rather than a refutation, since retrieval-side nDCG is neither necessary nor sufficient for the pedagogical-quality claim. The reframed H3 was supported: the calibrated LLM assessor (Claude Opus 4.8) reproduced both human-qrels retrieval verdicts (four-system ranking concordant on five of six pairs, Kendall's $\tau = +0.67$) and agreed with individual teachers above the mixed-expertise human ceiling ($\alpha = 0.59$ vs. 0.46), with a documented dense-channel tilt as its principal limitation (§4.5).

Contribution and honest scope. The thesis contributes pedagogical retrieval (the worked-example effect operationalised as a retrieval-time content-type selection policy); a domain-

transfer validation of the LLM-as-relevance-assessor method to this francophone, low-resource setting; and a documented, reusable Ivorian Terminale C corpus and data card, the first of its kind to the author's knowledge. It establishes the retrieval-and-evaluation foundation on which the generation-quality study is the immediate, fully-specified next step.

Preface and Acknowledgements

This research paper was completed for the MSc Data Science for Business (HEC Paris / École Polytechnique), academic year 2025–2026.

I thank my supervisor, Paul ASSALE, for his guidance throughout this work. Recruiting a curriculum-proficient annotation panel in a short window would not have been possible without the help of Audrey KIRE, Abigail ANANI, Yann OKOUA, Jacob ACCOH, Kenia BEAT, Laure YEO, Michele AGBESSI, and Nadjin OUATTARA, who connected me with the mathematics teachers and Terminale C students who provided the relevance judgements. Those annotators, who contributed under anonymised identifiers, are the foundation of the evaluation reported here, and I am grateful to them.

Use of AI tools. This work made deliberate and disclosed use of AI assistants. Anthropic's Claude (via the Claude.ai Projects interface) was used for research discussion, concept clarification, and drafting and refining the written report; Claude Code, run inside Visual Studio Code, was used as a coding assistant for the implementation, the experiments, and the build tooling. All research design, methodological decisions, analysis, and final claims are my own, and every AI output was reviewed and verified before inclusion.

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Chapter 1. Introduction

1.1 Motivation

In Francophone West Africa, the baccalauréat is the single high-stakes examination that gates access to higher education, and success rates are low. In Côte d'Ivoire specifically, the national BAC pass rate was 40.15% in 2025, with only 133,518 of 332,515 candidates admitted [APAnews, 2025]. It has stayed in that band across the past decade (roughly 29–46%; e.g. 32.09% in 2023) [Yeclo, 2023]. Within the baccalauréat, the série C track (the mathematics- and physical-science-intensive stream) is both the most selective and the most consequential for entry into engineering, medicine, and the quantitative disciplines.

Performance in Terminale C mathematics is therefore a meaningful determinant of a student's trajectory, but the resources a student can bring to bear on it are unevenly distributed. Qualified mathematics teachers are scarce (a shortage acute enough that a March 2026 exceptional state recruitment drew 7,668 candidates for 2,000 contractual mathematics and physical-science posts [Connectionivoirienne, 2026]), and access to them is better in urban centres than rural areas, while structured revision outside the classroom is hard to come by. The financially fortunate hire private tutors, but at a price out of reach for the majority: a well-known Abidjan tutor put competent tutoring at on the order of €100 per month [Le savant ACHI, personal interview, 2 February 2026], close to the Ivorian minimum monthly wage (the SMIG, about 75,000 FCFA ≈ €114 [Gouv.CI, 2023]). An Ivorian family thus spends nearly a full minimum wage on tutoring, where in France the same minimum wage (the SMIC, ≈ €1,800/month [Travail-Emploi.gouv.fr, 2026]) makes the relative burden far lighter. Good private tutoring is concentrated among families who can afford it and in the cities where tutors are present. This access asymmetry is the practical problem that motivates this thesis, whose aim is to give students affordable tools for self-study.

Large language models (LLMs) appear, at first sight, to offer a route to closing that gap: a conversational tutor available at any hour to any student with a connected device. But the naïve application of a general-purpose LLM to high-stakes mathematics tutoring is problematic in three specific ways that this thesis takes as central: two concerning what and how the model answers, and a third, compounding risk in the reliability of those answers.

First, a general-purpose model is not aligned to a specific curriculum. It may solve a problem by a method the student has not been taught, invoke a theorem outside the syllabus, or use notation and conventions that diverge from the official West African programme. For a Terminale C student preparing for a specific examination against a specific textbook, this curriculum drift is itself a failure mode, independent of mathematical correctness: a technically valid solution that

departs from the prescribed method is, for this student, a wrong answer. This is the gap that grounding is meant to close, and it is the principal motivation for the retrieval-based approach taken here.

Second, access to the most capable models is, in practice, rationed. The frontier models a student can reach for free impose daily usage limits that a serious revision session exhausts quickly; the models without such limits cost money the student does not have. The practical question this raises is whether a small, locally deployable model, cheap enough to run without per-query cost or hard usage caps, can be made good enough to serve as that tutor once it is properly grounded. This is the deployment question behind the generator comparison of H4.

A general-purpose model carries a further, well-documented risk that compounds the first: LLMs hallucinate: they produce fluent, confident statements that are factually or mathematically false [Huang et al., 2025]. This is a general property of how these models generate text, not a phenomenon specific to any one region or curriculum, and this thesis makes no claim about its measured prevalence among Ivorian students in particular. Its danger here is one of amplification: hallucination is most harmful precisely where the learner cannot readily verify the output, and a Terminale C student facing an unfamiliar derivation is, by definition, poorly placed to tell a correct one from a plausible-looking wrong one. In that setting, fluent error is more dangerous than visible failure, because it is more likely to be believed and carried forward. The thesis therefore treats faithfulness (the model's adherence to its grounded sources rather than to its own unconstrained generation).

Retrieval-augmented generation (RAG) [Lewis et al., 2020] is the natural first step against the first and third of these dangers. By grounding the model's output in retrieved passages from an authoritative curriculum source, RAG constrains the tutor to the syllabus and gives it a factual basis against which faithfulness can be enforced and audited. RAG is, moreover, an established hallucination-mitigation approach in the broader literature [Lewis et al., 2020; Huang et al., 2025]. The authoritative source in the Ivorian context is the CIAM series (Collection Inter-Africaine de Mathématiques), the canonical regional textbook aligned with the official Terminale C programme. This thesis builds and evaluates a RAG-based mathematics tutor grounded in the CIAM Terminale C corpus and, addressing the second danger, pairs that pipeline with a small language model (SLM) whose grounded response quality is compared against that of a frontier LLM of the kind students currently use, to test whether a low-cost, locally deployable tutor can approach frontier quality once both are grounded.

Beyond the immediate artefact, the work is intended to be decision-relevant for the people choosing whether and how to fund AI for education in the region. For entrepreneurs and government bodies weighing investment in AI-based learning tools (a space drawing growing public and private investment in the region, with national education budgets in the billions of euros and a nascent EdTech startup ecosystem [TeamFrance-Export, 2025], and with Côte d'Ivoire among the lead countries developing a national AI-in-education competency framework under the GPE-KIX/UNESCO initiative [GPE-KIX, 2025]), a grounded, curriculum-aligned evaluation provides evidence on whether such tools can deliver curriculum-faithful tutoring at acceptable quality, and at what cost, rather than leaving the question to intuition.

The francophone, curriculum-specific, low-resource character of this setting is not incidental. Most published work on LLM tutors and on retrieval for education targets English and draws

on English-language resources; the EdTech tools that reach West African students are predominantly anglophone and curriculum-agnostic. Building and, critically, evaluating a tutor for a francophone curriculum with scarce labelled is very rare and addressing this is part of the contribution of this work.

1.2 Problem statement

This thesis is organised around three gaps between what general-purpose RAG offers and what a mathematics-focused francophone West African curriculum tutor requires. A fourth, lighter question (whether a grounded small model can stand in for a frontier model at acceptable quality and cost) is opened rather than closed by this work, and is treated as the deployment question behind H4 (§1.3) rather than as a fully resolved gap.

Gap 1: generic retrieval is content-type-blind. Standard retrieval, whether sparse, dense, or a hybrid of the two, ranks candidate passages by a single relevance signal and is indifferent to what kind of content it returns. But solving a Terminale C problem requires the student to mobilise both the underlying theory (a definition, theorem, or method) and a worked example demonstrating its application; the pedagogical value of pairing the two is well established [Sweller, 1988]. A content-type-blind retriever can return five theory passages and no worked example, or vice versa, and either failure degrades the tutor's pedagogical utility. Closing this gap requires retrieval that is aware of, and enforces, a pedagogically motivated content-type mix.

Gap 2: LLM-as-judge evaluation is rarely calibrated against domain experts. Evaluating a tutor's free-text responses at scale is infeasible by human annotation alone, and the field has increasingly turned to using a strong LLM as an automated judge [Zheng et al., 2023]. But an uncalibrated LLM judge risks two distinct failures in this education setting. It may simply be unreliable, disagreeing with expert human judgement to a degree that makes its scores uninformative. And it may be circular: if the same family of model both generates and judges, and if "quality" is operationalised as "what the judge model finds good," then the evaluation rewards the generator for sharing the judge's biases rather than for tutoring well, and any comparison of generators against that judge is compromised. Closing this gap requires a judge whose agreement with domain-expert human judgement is demonstrated rather than assumed, and an evaluation framework whose notion of quality is anchored to that human judgement rather than to model preference.

Gap 3: the francophone, low-resource, curriculum-specific setting is under-studied. The methods above (hybrid retrieval, pedagogical retrieval, calibrated LLM-judging) have to be made to work in French, over OCR'd textbook content dense with mathematical notation, with a labelling budget of a small curriculum-proficient panel rather than a crowd. Each of these conditions interacts with the methods in ways that the anglophone, more data-rich literature does not address, and documenting those interactions, including where they constrain what can be concluded, is part of the problem this thesis takes on.

1.3 Research questions and hypotheses

The umbrella research question of this thesis is:

> What form of retrieval-augmented generation can be adapted to the pedagogical structure of a high-stakes mathematics curriculum, and how can the resulting tutor be evaluated rigorously when domain-expert annotation is scarce and the targeted population is low-resource?

This decomposes into four sub-questions, each operationalised as a testable hypothesis. The hypotheses are stated here and developed in full in the methodology of Chapter 3.

- RQ1 / H1 (retrieval channel). Does fusing sparse (BM25) and dense (multilingual) retrieval improve retrieval quality over dense-only retrieval for francophone curriculum mathematics? H1: a reciprocal-rank-fusion hybrid outperforms dense-only retrieval on standard ranking metrics against the human-panel relevance judgements (Chapter 3).
- RQ2 / H2 (pedagogical retrieval, a central contribution). Does enforcing a pedagogically motivated content-type mix (theory passages together with worked examples) at retrieval time improve tutoring quality beyond content-type-blind hybrid retrieval? H2: a pedagogical retriever that selects a fixed mix of theory-type and example-type passages from the hybrid candidate pool outperforms the content-type-blind hybrid baseline.
- RQ3 / H3 (evaluation calibration). When expert labels are scarce, can an LLM be calibrated to act as a relevance assessor, reproducing the human panel's (query, chunk) relevance judgements faithfully enough to substitute for them at scale? H3: an LLM relevance assessor, prompted with the same rubric the panel used and applied to the (query, chunk) pairs, (i) produces qrels that rank the four retrievers (dense, BM25, hybrid, pedagogical) in the same order as the human-panel qrels, and (ii) reaches per-item agreement with held-out annotators (Krippendorff's α) that approaches the human inter-annotator agreement ceiling, rather than meeting a fixed numerical threshold. The ceiling is that of a mixed-expertise panel (two teachers and eight Terminale C students; Chapter 3), and is read in that light.
- RQ4 / H4 (deployment feasibility). When properly grounded, do open-weight small language models close the quality gap with frontier models, to a degree that would make low-cost local deployment feasible? H4: under grounding, the frontier–open quality gap (measured as a difference-in-differences against the ungrounded gap) narrows substantially.

The status of each hypothesis differs, and this thesis is explicit about that difference throughout (§1.4, and the BUILT/PLANNED convention of Chapter 3). H1 and H2 (retrieval side) are tested end-to-end: the (query, chunk) relevance-annotation package was dispatched, returned, and scored, and the populated results are reported in Chapter 4. H3's human inter-annotator ceiling is computed from the overlap-annotated anchor, and the reframed H3 relevance assessor that it validates was run on those same returned labels and is likewise reported. The original formulation of H3 (an LLM grading full tutoring responses on a four-criterion rubric) is reframed to future work (and delegated externally as a no-cost freeroll), because it rests on a separate response-grading annotation campaign which responses took more time to collect before submission of this report. H4 is fully specified but not entirely executed (the human calibration is pending).

1.4 Contributions

This thesis makes four contributions. They are stated here with explicit attention to what has been executed versus what has been specified but not executed at the time of writing, so that the scope of each claim is unambiguous from the outset.

C1 — Pedagogical retrieval (novel method; H2). The principal methodological contribution is pedagogical retrieval: a retrieval-time selection policy that enforces a pedagogically motivated mix of theory-type and worked-example-type passages, drawn from the same fused candidate pool as a standard hybrid retriever and differing from it only in selection. Its retrieval-side evaluation against the human-panel relevance judgements was executed end-to-end, and the populated results are reported in Chapter 4 (§4.3). Its generation-side evaluation (via the response judge of C2 and the student study of Chapter 3) is specified but not yet executed.

C2 — A human-calibrated, model-independent evaluation framework, and a validated LLM relevance assessor (H3). The second contribution is an evaluation framework built around a relevance rubric whose notion of quality is anchored to a correct human solution and is deliberately model-independent, avoiding the circularity that would arise from defining relevance as "what an LLM finds useful." The framework comprises the rubric, the annotation protocol with its overlap design, and the agreement machinery (Krippendorff's α, qrels construction, ranking metrics); the rubric design, the annotation tooling, and a two-teacher calibration pilot are executed, the pilot having driven a documented revision of the rubric (v0.1 $\rightarrow$ v1.0), and the full dispatch, to a mixed panel of two teachers and eight Terminale C students, has been sent and returned. On this foundation, H3 asks whether an LLM relevance assessor (the same model, prompted with the panel's rubric) can reproduce the panel's (query, chunk) judgements well enough to stand in for them: validated primarily by whether its qrels rank the retrieval systems as the human-panel qrels do, and secondarily by per-item agreement approaching the human inter-annotator ceiling rather than a fixed α bar. This adopts an established, independently reproduced IR-evaluation method [Faggioli et al., 2023; Thomas et al., 2024; Upadhyay et al., 2024] and contributes its domain-transfer validation to a francophone, low-resource, curriculum-mathematics setting, calibrated against a curriculum-proficient human panel rather than crowd workers.

C3 — A documented, reproducible Ivorian Terminale C corpus and data card (artifact). The third contribution is a concrete, reusable research artifact: a chunked, content-type-tagged corpus of 722 chunks drawn from ten CIAM Terminale C chapters and two BAC série C papers, with a full data card reporting corpus statistics, OCR-quality validation, a documented eight-value content-type taxonomy, and an explicit account of the corpus's limitations (figure dropping, geometry exclusion, a quantified worked-example tagging hole, and leakage discipline via held-out BAC 2025). To the author's knowledge this is the first openly documented RAG corpus for the Ivorian Terminale C mathematics curriculum.

C4 — A specification of a grounded frontier-versus-open generator comparison (H4). The fourth contribution is a fully specified experimental design for testing whether grounding closes the quality gap between open-weight small models and frontier models, framed as a difference-in-differences against the ungrounded gap, with the metric, the grounding manipulation, and the analysis pre-specified.

1.5 Scope and non-claims

The boundaries of this work are stated up front.

- Geometry is out of scope. The corpus comprises ten non-geometry chapters of the CIAM Terminale C textbook; geometry chapters are excluded because their constitutive dependence on figures is not fully preserved through the OCR pipeline (Chapter 3, §M3.4). All claims, including H1 and H2, are claims about Terminale C content excluding geometry.
- A single official textbook. All curricular content is drawn from the official CIAM textbook series. Third-party textbooks that many students use for revision outside class are not represented, and generalisation to them is not directly supported by the experiments here.
- The tutor is single-turn. The generation pipeline answers a single question against retrieved context; multi-turn dialogue is not implemented and is identified as future work.

Stating these limits at the outset is deliberate. A central aim of the thesis is to demonstrate methodological discipline under real constraints (scarce expert labels, OCR'd low-resource content, a hard deadline), and an honest account of what the work does not establish is part of that demonstration.

Chapter 2. Related Work

2.1 Retrieval-augmented generation and hybrid retrieval

Retrieval-augmented generation (RAG) couples a parametric language model with a non-parametric retrieval step: at inference the model is conditioned on passages fetched from an external corpus, so its output is grounded in retrieved evidence rather than parametric memory alone (Lewis et al., 2020). Originally motivated by knowledge-intensive tasks, the same mechanism has become a standard response to hallucination, because grounding gives the generator a verifiable source against which its claims can be constrained and audited (Huang et al., 2025). This thesis adopts RAG for that second reason — curriculum fidelity and faithfulness, not open-domain knowledge coverage.

The retrieval step admits a well-known design space. Sparse retrieval scores a query against documents by lexical overlap, canonically BM25 (Robertson & Zaragoza, 2009). Dense retrieval embeds query and document into a shared vector space and ranks by similarity, capturing semantic relatedness that survives paraphrase and vocabulary mismatch (Karpukhin et al., 2020). The two have complementary failure modes: sparse retrieval is precise on rare tokens — names, notation, theorem labels — but brittle to wording; dense retrieval is robust to wording but can wash out the exact-match signal mathematical content depends on. A hybrid retriever combines the two rankings, commonly by Reciprocal Rank Fusion (RRF), which requires no score calibration between systems (Cormack et al., 2009). That fusion outperforms either channel alone, on a French OCR'd mathematics corpus dense with notation, is precisely what H1 tests.

Two conditions of the present setting are under-represented in this literature. First, almost all of this work is benchmarked on English, whereas this corpus is French; the dense channel here relies on a multilingual sentence-embedding model (multilingual E5; Wang et al., 2024), selected for French retrieval quality. Second, retrieval for education — over curriculum material, evaluated for pedagogical rather than topical relevance — is a comparatively thin literature. H1 therefore does not propose a new retrieval mechanism; it asks whether an established one transfers to a francophone, low-resource, notation-heavy curriculum corpus — an answer the English-language results do not settle, because the very signal hybridisation recovers (exact lexical match on notation and theorem names) behaves differently in OCR'd mathematics than in clean web text.

2.2 The worked-example effect and pedagogically structured retrieval

The second strand is an instructional-design literature, and it motivates the thesis's central methodological contribution. Cognitive Load Theory holds that learning is constrained by the limited capacity of working memory, and that formats which reduce extraneous load free capacity for the schema construction that constitutes learning (Sweller, 1988). A repeatedly replicated finding is the worked-example effect: for novices acquiring a problem-solving skill, studying worked examples — a problem paired with a fully elaborated solution — produces better and more efficient learning than solving the equivalent problems unaided (Atkinson et al., 2000). The pedagogical unit that supports a novice is therefore not theory alone and not an example alone, but the pairing: the governing principle (definition, theorem, or method) together with a demonstration of its application.

This is the premise behind H2. A Terminale C student working a problem must mobilise both the underlying theory and a worked example of its use; a content-type-blind retriever can return five theory passages and no example, or the reverse, degrading the tutor's instructional value independently of each passage's topical relevance. The thesis's response — pedagogical retrieval — is a retrieval-time selection policy that enforces a fixed mix of theory-type and worked-example-type passages from the same fused pool as the hybrid baseline, differing only in what kind of content fills the context window.

To the author's knowledge, the worked-example effect has not previously been operationalised as a retrieval constraint. The instructional-design literature studies how to present worked examples to a learner who already has them; the IR literature optimises topical relevance and is indifferent to pedagogical content type. Pedagogical retrieval sits in that gap — importing a result from cognitive science into the selection stage of a RAG pipeline. This is the sense in which C1 is novel: not a new retrieval algorithm, but a content-type-aware selection policy whose objective is pedagogical composition rather than relevance maximisation alone. The claim is one of specific novelty, not of an empty field: educational RAG is active — surveyed by Li et al. (2025) — and recent systems adapt retrieval to the learner. The closest neighbour, PRAG-EDU (Nguyen et al., 2026), conditions retrieval and response complexity on a student's historical grades as a proficiency signal — a different axis from the content-type composition proposed here. To our knowledge, no prior method enforces a fixed theory-plus-worked-example content-type mix at the selection stage of a curriculum-tutoring RAG pipeline.

The method is called pedagogical retrieval rather than instructional retrieval (the codebase name) to avoid collision with the distinct instruction-following retrieval literature, where "instruction" denotes a natural-language directive attached to a query, not a pedagogical worked example.

2.3 LLM-based relevance assessment for retrieval evaluation

Evaluating retrieval requires relevance judgements: to compute Recall@k or nDCG@k one must know which chunks are relevant to which query, and those labels are the answer key against

which ranking is scored. Producing them at scale is the classic bottleneck of IR evaluation, and a fast-moving literature asks whether an LLM can produce them in place of human assessors.

Faggioli et al. (2023) frame a spectrum of human–machine collaboration in relevance judgement and raise the circularity objection explicitly — that using an LLM to judge relevance may unfairly reward systems that themselves rely on LLM-like semantic similarity. Thomas et al. (2024), from a major commercial search team, supply the method this work most closely follows: take human relevance feedback, search over LLM-and-prompt configurations for one whose judgements agree with it, then label at scale; they report LLM assessors about as accurate as human labellers at predicting searcher preferences, while cautioning that agreement is sensitive to prompt phrasing and that LLM labels can be biased toward rankers incorporating embedding similarity. Upadhyay et al. (2024) provide UMBRELA, an open reproduction built on an openly available frontier model, used for the TREC 2024 RAG Track, with a companion large-scale study of LLM-assessment properties. Together these establish LLM-as-relevance-assessor as an established and independently reproduced method.

This is the literature the reframed H3 is anchored to. H3 as executed does not propose LLM relevance assessment — that method exists and has been reproduced. The contribution is a domain-transfer validation: the method is carried into a setting the source literature does not cover (francophone, low-resource, curriculum mathematics) and validated against a curriculum-proficient human panel — two practising mathematics teachers with former and current Terminale C students — rather than crowd or searcher labels, with the success criterion set by the human agreement ceiling rather than a fixed threshold. Two lessons are imported directly: the assessor is given the same formalised information need the panel used (the rubric), following the finding that formalised topics improve human–LLM agreement, with the prompt and decoding locked and paraphrase sensitivity recorded as a limitation; and the headline validation is system-ranking agreement — whether LLM labels rank the four retrievers in the same order as the human labels — the field's primary validation mode (Thomas et al.; UMBRELA), with per-item agreement against the human ceiling reported as secondary.

The circularity concern is contained by design: because the retriever verdicts (H1, H2 retrieval side) are computed on the human-panel qrels and never on LLM qrels, the assessor is the object under test, never the source of truth for the retrieval conclusions. The dense/semantic-similarity tilt Thomas et al. warn of is therefore recorded as a limitation of the assessor, not a contaminant of H1/H2.

2.4 LLM-as-judge for open-ended response evaluation

A second, larger LLM-evaluation literature concerns grading open-ended generated responses, and keeping it distinct from §2.3 is the methodological hinge of this thesis's evaluation design. The canonical reference is Zheng et al. (2023), which studies using a strong LLM to judge chatbot answers (MT-Bench) and pairwise preference battles (Chatbot Arena), reporting that a capable LLM judge agrees with human preferences about as often as humans agree with one another. It also documents the failure modes that make naïve LLM judging risky — position bias, verbosity bias, and, most consequentially for a generator comparison, self-enhancement bias, the tendency to favour outputs from the judge's own model family.

The two literatures differ in their annotation object, and conflating them was the error the H3 reframe corrected. Relevance assessment (§2.3) labels a (query, chunk) pair on a relevance scale; the judgement is front-loaded into an answer key and subsequent scoring is mechanical. Response judging grades a (query, full response) pair on a multi-criterion rubric — factual accuracy, curriculum alignment, pedagogical quality, language register — and is itself the open-ended judgement. A relevance label on a chunk says nothing about whether a generated answer tutors well. The thesis's executed evaluation result is the relevance assessor of §2.3; the four-criterion response judge is specified but not executed, documented as future work (H3b) alongside the generation-side test of H2 and the H4 comparison it would have scored. The deferral is structural: the response judge depends on a (query, response) gold set the teacher dispatch never produced (only the (query, chunk) relevance package was sent and returned).

Self-enhancement bias is why the future judging framework is designed to be model-independent — anchoring quality to a correct human solution and keeping any judge model outside the set of generators it scores, so the H4 comparison cannot be won merely by sharing the judge's family preferences (C2). That this circularity is weaker for relevance assessment than for response judging is one more reason the reframed H3 was chosen as the executable result.

2.5 Positioning: the gap and this thesis's place in it

The four strands meet at a setting none individually covers. RAG and hybrid retrieval (§2.1) are mature but developed on English, data-rich, open-domain text; the worked-example effect (§2.2) is a robust pedagogical result not turned into a retrieval constraint; LLM relevance assessment (§2.3) and response judging (§2.4) are established methods validated against crowd or general-preference labels in English. This thesis assembles these strands for a francophone, low-resource, curriculum-specific secondary-school mathematics tutor, evaluated with a small curriculum-proficient panel (two teachers and eight former or current Terminale C students) over OCR'd notation-dense content — a combination the source literatures do not jointly address.

The contributions locate against that gap as follows. C1 — pedagogical retrieval (H2) is the one genuinely new method: the worked-example effect operationalised as a content-type selection policy at retrieval time. C2 — a human-calibrated, model-independent evaluation framework (H3) adapts the relevance-assessment method of §2.3 to a curriculum-proficient panel and a ceiling-based criterion, and specifies the response-judge extension of §2.4 with its circularity safeguards. C3 — the documented Ivorian Terminale C corpus and data card is a reusable artifact for a curriculum these literatures have not served. C4 — the grounded frontier-versus-SLM specification (H4) frames the deployment question they leave open for low-resource settings.

Consistent with the honesty discipline maintained throughout, the executed core is H1, the retrieval side of H2, and the reframed H3 — three results resting on the corpus and the relevance labels already in hand. The response judge of §2.4, the generation side of H2, and H4 in full are specified and anchored to the literatures surveyed here, and carried as future work rather than claimed as results.

Chapter 3. Methodology

M1–M4. Corpus, chunking, statistics, and retrieval

M1. Corpus construction

Source. The corpus is the CIAM Terminale C series (Collection Inter-Africaine de Mathématiques, ed. Saliou Touré), the official Ivorian Terminale C textbook. Ten non-geometry chapters were indexed alongside two official BAC série C papers (2022, 2024), for twelve documents (full list, data card §2.2); geometry is excluded. The BAC série C 2025 paper was held out from indexing: nine of its items appear in the evaluation set, so quarantining it prevents the retriever from surfacing a gold-standard solution as a retrieved passage.

OCR and cleaning. Scanned PDFs were converted to text-plus-LaTeX with `marker` [Paruchuri, 2024], chosen for being locally executable, open-source, and structure-preserving (headings and inline-LaTeX equations), supporting both the content-type taxonomy (§M2) and equation-aware metrics. Validation on one hand-transcribed chapter gives prose CER 11.51% / equation exact-match 87.46% (full figures, Annex E), typical for equation-dense textbook OCR, and the precondition for the downstream generation and faithfulness part in §M6; the single-chapter scope makes it indicative, not definitive. The marker model presented degeneration in multiple independent runs of the OCR. Thus, post-processing dropped figure regions and excised vision-model repetition loops, leaving a sentinel in their place. Six syntactic checks then set a per-chunk hallucination flag: five heuristics for likely OCR damage, plus the sentinel test, whose presence is proof rather than suspicion that content was lost. Flagged chunks are retained but tracked, so a downstream failure on one is distinguishable from a genuine retrieval failure. (retained but tracked: rates in §M3.1; `src/ocr/clean.py` and `src/corpus/filters.py`, data card §4–5)

M2. Chunking and the content-type taxonomy

The cleaned corpus was segmented based on a specific structure organised in units. So, each chunk is one of the 8 typographic units: theory block, theorem, proof, worked example, definition, exercise, solution, and remark. There was no uniformed size / length cap for each chunk, meaning that even two different theory chunks could have different sizes. Also, empty-body headings are dropped. After all that segmentation of the clean corpus, the corpus obtained consisted of 722 chunks. Each carries exactly one of eight labels assigned by a deterministic keyword heuristic over marker's content heading (and not topic heading) text. The code to identify the content heading and assign the label can be seen in (`src/corpus/chunking.py`). One limitation is load-bearing for H2: some worked examples are not titled "examples" in the

book. Hence they are usually silently labelled `theory`, contaminating the theory bucket. This was sorted to some extent, as can be seen in §M3.3 and §M5.5.

M3. Corpus statistics, evaluation set, and limitations

M3.1 Corpus statistics

722 chunks; 424,010 prose characters; 75,853 words; 5,313 equations; 15.7% of chunks carry $\geq$ 1 dropped figure region and 14.7% a hallucination flag. Per content type: theory 185 (25.6%), theorem 142 (19.7%), example 141 (19.5%), remark 82 (11.4%), proof 57 (7.9%), definition 48 (6.6%), exercise 34 (4.7%), solution 33 (4.6%). The pedagogical retriever's (§M5.2) bucket aggregates are thus 514 theory-bucket (theory + theorem + definition + proof + remark), 174 example-bucket (example + solution), and 34 excluded `exercise` ($\approx$ 3:1). The chunks labelled "exercises" were excluded because they do not have any solution given in the book. Per-chapter composition is in Annex D.

M3.2 Evaluation query set

This is to evaluate the retriever. It consists of: 100 queries, stratified across the ten chapters and three structural categories — 60 single-leçon, 20 multi-notion, 20 hard-edge. By source, 91 synthesised / 9 BAC-2025: the 9 BAC-2025 queries come from the de-indexed 2025 paper and give an ecologically valid slice of authentic problems, while the synthesised queries supply stratified coverage and the multi-notion/hard-edge density that BAC items alone cannot (authoring protocol, §M7).

M3.3 Tagging-quality estimates

Tagging quality rests on two targeted audits (data card). A precision check: of the 24 worked examples recovered from `theory`-labelled chunks by the imperative/study cues, all 24 were genuine worked content (zero false sweeps). A recall check surfaces the dominant residual error: a manual audit of n = 40 `theory`-labelled chunks (seed 42) estimates $\approx$ 32% are in fact headingless worked examples (95% CI 20–48%). The dominant tagging error is therefore one-directional `example`→`theory` leakage of unmarked content, not cross-class confusion; its H2 consequences are carried in §M5.5 and full-corpus quantification is future work (§M9).

M3.4 Geometry exclusion

Geometry chapters and geometry queries are excluded: geometry is densely figure-dependent, so indexing figure-stripped geometry chunks would yield text-only passages with systematically lower retrievability and pedagogical utility than analytic content, confounding any cross-chapter comparison. A figure-preserving multimodal pipeline is the cleaner long-term solution and is future work (§6.3).

M3.5 Other limitations

Figure-dropping (15.7%) means a failure on a figure-dependent passage cannot be cleanly attributed to a retriever or generator defect; hallucination-flagged chunks (14.7%) are tracked

separately so a failure on them is distinguishable; all retrieval experiments report the 91 synthesised and 9 BAC-2025 queries separately, so any difference between authored and authentic queries is visible rather than averaged away; and generalisation to third-party Terminale C textbooks is not directly supported (single-textbook corpus).

M4 Retrieval:

M4.1 BM25, dense, and the RRF hybrid

The stack for the retrieval has three channels: a keyword-matching, a semantic-matching, and a mix of both
- Sparse BM25 [Robertson & Zaragoza, 2009] over a French-tokenised corpus - Dense multilingual-E5 [Wang et al., 2024] (1024-dim, documented French coverage) ranked by cosine similarity. - RRF hybrid [Cormack et al., 2009] fusing the two. The hybrid is the baseline against which the pedagogical retriever (§M5) is contrasted under H2; hybrid-vs-dense is the registered test of H1 (§M4.4).

The hybrid (and the later explained pedagogical retriever in M5) consume the same RRF-fused pool with each of BM25 and dense contributing to its top-20, re-ranked by $\mathrm{rrf}(d) = \sum_r 1/(60 + \mathrm{rank}_r(d))$. Its top-5 in fused order is the hybrid's output.

The maths-motivated BM25 tokenisation (single-character-variable preservation, accent folding, no lemmatisation), the E5 `query:`/`passage:` prefix convention, and the fusion parameters are detailed in Annex M.E.

M4.2 H1 evaluation protocol and status

H1 tests whether the RRF hybrid outperforms dense-only on the 100-query set against the human-panel qrels (§M7). Three metric families are reported at $k \in \{5, 10\}$ — Recall@k (binary, ≥ 2), nDCG@k (graded 0–3), MRR — with primary metric nDCG@5, paired-bootstrap significance (§M9), and secondary analyses by source (leakage check) and query type. The retrieval infrastructure is built; the full (query, chunk) relevance package of 1,565 pairs was dispatched, returned, and processed into qrels.
The H1 results are reported in Chapter 4 and state that the hybrid's advantage over dense comes from the keyword channel, and it holds only on the 91 synthesised queries while the ordering reverses for the 9 authentic BAC items. Because the synthesised queries were written in the textbook's own vocabulary, part of that advantage may be an artefact of how they were constructed rather than a property of the retriever.

M5. Pedagogical retrieval

This relates to Hypothesis 2. ### M5.1 Motivation

The retrievers of §M4 are content-type-blind: they rank by relevance signals that have no constraint on what kind of content (theory, or definition, example, etc) is returned. For a Terminale C student, this approach might be judged irrelevant because they will need the information given to them to be structured. Solving a typical problem requires both the underlying machinery (a definition, theorem, or method) and an example showing it applied. This has been

documented as the worked-example effect [Sweller, 1988; Atkinson et al., 2000] and is mirrored in CIAM (the offficial curriculum texbook) itself, where each leçon interleaves exposition with "exercices résolus". A blind retriever's top-k may be almost all theory or almost all examples; either failure mode degrades tutoring (theory alone leaves no applied template; examples alone leave no formal scaffolding to generalise).

The pedagogical retriever is therefore a selection policy that enforces a content-type mix at retrieval time: a fixed number of theory passages and a fixed number of example passages, both drawn from the same RRF-fused pool the hybrid baseline uses.

M5.2 Design choices

For slot allocation of that selection policy, the eight-value taxonomy used in the CIAM textbook collapses into two buckets and one excluded category:

Bucket	Constituent labels
Theory	`theory, theorem, definition, proof, remark`
Example	`example, solution`
Excluded	`exercise`

The theory bucket aggregates content that establishes formal claims or supplies motivating commentary; the example bucket aggregates content that show worked examples internal to a lesson (`example`) and the textbook's own `solution`s. `exercise` in the textbook are unsolved problems, with no guidance on how to solve them. Therefore, it has been deliberately excluded from both: a problem statement carries no instructional content.

Hence, the corpus segmentation for the pedogical retriever is: 514 theory-bucket, 174 example-bucket, and 34 excluded ($\approx$ 3:1).

The 3-theory : 2-example split is fixed for every experiment and is methodological, not tuned on a development set. Tuning $(k_{\text{theory}}, k_{\text{example}})$ on held-out data would convert H2 from a hypothesis test into a hyperparameter search, inviting multiple-comparisons concerns across the grid and, more substantively, answering the wrong question. The meaningful question is not whether some split outperforms the hybrid baseline but whether the principle of enforced theory–example mixing does at all. Fixing the split at a defensible a priori value (three theory passages, in approximate proportion to the dominant theory-bucket share) tests the principle without incidental-tuning noise. Treating the split as a tunable hyperparameter and reporting the resulting Pareto frontier is future work (§6.3).

As a fallback, it was decided that if the pool holds fewer than three theory-bucket or two example-bucket passages, backfill the unfilled slots with the highest-ranked remaining passages of either bucket, recording a per-query fallback flag used in §M5.5 to condition the H2 contrast on example-availability.

M5.3 The slot-allocation Algorithm

For each query, the pedagogical retriever proceeds in three steps.

(1) Pool construction with no re-ranking beyond RRF: the same RRF fusion used by the hybrid baseline of §M4 (each of BM25 and dense contributes its top-20; fused by $\text{rrf}(d) = \sum_r 1/(60 + \text{rank}_r(d))$ [Cormack et al., 2009]).

(2) Slot filling: traverse the fused list in descending order, assigning the first three theory-bucket passages to the theory slots and the first two example-bucket passages to the example slots, skipping `exercise`.

(3) Fallback: if the pool holds fewer than three theory-bucket or two example-bucket passages, backfill the unfilled slots with the highest-ranked remaining passages of either bucket, recording a per-query fallback flag used in §M5.5 to condition the H2 contrast on example-availability.

The five-passage payload is numbered `[1], …, [5]` in fused-rank order (preserving rank for the generator's citation behaviour, §M6). Production code is `src/retrieval/instructional.py` (instructional = pedagogical retriever)and here is a reduced reference version:

```
def pedagogical_retrieve(query, k_theory=3, k_example=2):
    pool = rrf_fuse(bm25.top(query, n=20), dense.top(query, n=20), k=60)
    theory, example, leftover = [], [], []
    for chunk in pool:                                  # descending RRF score
        if chunk.type in THEORY_BUCKET and len(theory) < k_theory:
            theory.append(chunk)
        elif chunk.type in EXAMPLE_BUCKET and len(example) < k_example:
            example.append(chunk)
        elif chunk.type != "exercise":
            leftover.append(chunk)
    fallback = (len(theory) < k_theory) or (len(example) < k_example)
    while len(theory) + len(example) < (k_theory + k_example) and leftover:
        theory.append(leftover.pop(0))                  # backfill, either bucket
    return theory + example, fallback
```

M5.4 How H2 is tested

H2 is therefore a hypothesis about selection, not ranking: baseline and pedagogical retriever see an identical fused pool and differ only in which five passages they select at fixed top-k = 5. In fact, there are 5 passages that would populate the generator's context window (§M6), although the H2 test reported in this thesis is purely retrieval-side and invokes no generator. Because the pedagogical retriever consumes the same RRF-fused pool as the hybrid baseline and returns the same number of passages, the H2 comparison focuses on the selection policy: any difference at fixed top-5 is attributable to which five passages are selected from a shared pool, not to tokenisation, embedding, fusion parameters, or pool size.

The pedagogical retriever is evaluated on the retrieval side against the human-panel qrels (§M7) under rubric v1.0. The primary retrieval-side metric is the per-query nDCG@5 difference between the pedagogical and hybrid retrievers over the 100-query set, with paired-bootstrap significance (§M9). Crucially, nDCG@5 is neither necessary nor sufficient for the H2 pedagogical-quality claim: the retriever deliberately passes over higher-relevance theory chunks to seat the example slots, so it can lose on nDCG@5 and still tutor better, or win with no tutoring gain.

The retrieval-side contrast therefore tests only whether the enforced mix is achieved without sacrificing relevance; the pedagogical-quality verdict itself is pending the generation-side evaluation (§M10). The infrastructure is built and was validated end-to-end on the 1,565-pair package; populated results include the per-query nDCG@5 and Recall@5 differences with bootstrap CIs, the example-slot fallback rate, and the example-availability-conditioned slice.

M5.5 Limitations

Two limitations are flagged here and revisited under threats to validity (§M9).

Baseline contamination from the theory bucket. A non-trivial fraction of `theory`-labelled chunks are in fact headingless worked examples — content demonstrating application that lacked the typographic marker the tagger keys on. A manual audit (n = 40 chunks read by hand, seed 42, 2026-06-10; data card §5.6) estimated ≈ 32% of `theory`-labelled chunks are headingless worked examples (95% CI 20–48%), about 60 chunks or 12% of the 514-chunk theory bucket (the other four theory labels were not audited); a full-corpus estimate is future work. To that extent, the hybrid baseline can already surface worked-example content through the theory channel, partially closing the gap the pedagogical retriever enforces. The H2 contrast could therefore be conservatively biased: any positive effect appears in spite of a baseline that captures, by chance, some of what the pedagogical retriever imposes by construction.

For the example-poor queries and the fallback regime, the fallback rate is reported with the H2 results. H2 is therefore reported in two slices: a marginal slice over all queries (the realistic estimate of operational impact) and a conditional slice restricted to queries that triggered no example-slot fallback (the cleaner test of the principle).

M6. Generation and faithfulness

Generation. The tutor answers a single question against the retrieved context using a frontier LLM. The generator is Claude Sonnet 4.6, called once per question (no multi-turn, no tool use), with decoding fixed at temperature 0 (`src/generation/client.py`). The five retrieved passages are passed in full, numbered `[1]…[5]` in fused-rank order (`src/generation/prompts.py`); no per-chunk truncation is applied, as even the longest chunks fit comfortably within the context window. The generator is built and its prompt was chosen by the controlled comparison below, but no answer to an evaluation query is generated or evaluated in this thesis; that evaluation is specified in §M10.

The V2 strict-faithfulness prompt. Because the corpus is OCR'd and ~15% of chunks carry a hallucination flag (§M3), the generator must not quietly "fix" its sources. The adopted prompt (V2) tells the model to report the retrieved extracts faithfully, even when they look mathematically wrong, and to raise any suspected error in a separate note, never to silently correct the source. This preserves curriculum authority (the student is being prepared against a specific textbook, §1.1) while still surfacing likely OCR errors for human review.

V2 was selected over a more lenient prompt (V1) and a stricter one (V3, which forbids any comment) in a controlled comparison run under the generator's actual decoding (Sonnet 4.6, temperature 0, 1,024-token cap). Each prompt answered three single-extract questions, three times each (27 calls): an extract containing an error the model can recognise (the textbook's

$|b| \leq |a|$ changed to the false $|b| \geq |a|$), one containing an error it cannot recognise (a tile size in a BAC exercise changed from 25 cm to 24 cm), and an unaltered extract (the definition of divisibility). The pass rule was fixed before running: report the extract and flag the error on the first; report the extract on the second; answer correctly without a false alarm on the third.

Extract	V1 (lenient)	V2 (strict)	V3 (hard)
Error the model can recognise	overrides the source and attributes its correction to [1]; no flag (0/3)	reports the source, flags the error (3/3)	reports the source, no flag (0/3)
Error it cannot recognise	reports the source (3/3)	reports the source (3/3)	reports the source (3/3)
Unaltered extract	correct, no false alarm (3/3)	correct, no false alarm (3/3)	correct, no false alarm (3/3)

V2 is the only prompt that passes all three (`experiments/2026-10-21-prompt-ablation-temp0/`; this supersedes an earlier single-run probe at the API's default temperature, `experiments/2026-06-01-bas`). With three items, this checks that the prompt behaves as designed; it is not a quantitative estimate of faithfulness.

A known limitation follows directly from the second row: no prompt flagged the error the model had no means to recognise. V2 can only flag errors the model has prior knowledge to detect; an error in content it has not seen, such as a wrong number in an exercise or a mis-OCR'd coefficient, passes through silently (§5.6). Two smaller observations: V2 does not always obey its own rule against adding examples (one of three answers on the unaltered extract added a worked example; V1 did so in all three), and no answer approached the 1,024-token cap (longest: 275 tokens), though these short questions do not settle whether long worked solutions would be truncated.

M7. Relevance rubric and annotation protocol

This section specifies the instrument behind the retrieval ground truth: the rubric, the calibration pilot that revised it, the overlap design, and the dispatch under which labels were collected. Agreement statistics are defined in §M9; the full rubric and protocol minutiae are in Annexes A and M.

M7.1 The annotation object

The unit is a (student question, retrieved passage) pair: a rater grades how relevant one retrieved passage/chunk is to answering the query. These labels constitute the qrels (query relevance judgements) i.e: the human answer key against which Recall@k and nDCG@k are computed to score each retriever by checking the chunks it returned. Those scores are what test H1 and the retrieval side of H2. After the H3 reframe the labels are also the gold against which the LLM relevance-assessor (§M8) is validated — which is why the definition below is engineered to be model-independent.

M7.2 Design choices: The rubric

M7.2.1 relevance definition: utility for a correct solution

The rubric anchors relevance to utility for a correct answer, not topical similarity. Hence for an annotator to judge if a pair is relevant or not, they ask themselves the **guiding question**: "To answer the question correctly, would a written solution use this retrieved passage?" (FR: « …une solution rédigée utiliserait-elle ce passage ? »). Therefore, relevance is judged by a curriculum-proficient human, independent of any model, not by what an LLM would find useful. A model-preference gold would be unjudgeable for human annotators and would contaminate the two comparisons the thesis must keep clean: validating the assessor against independent human judgement (§M8) and the future frontier-vs-SLM comparison. This is the C2 model-independence principle (§1.4; `decisions/2026-06-18-rubric-v1.0.md`), relied on throughout §M8.

M7.2.2 The 0–3 scale

As "relevance" has been defined, each pair receives a graded relevance score:

Score	Meaning (gloss)
3 — Sufficient	The passage's information alone suffices to answer all parts of the question.
2 — Useful	A part of the passage clearly helps with a part of the question — including a required method/format (e.g. an empty variation table for a function study), or a same-method worked example with different numbers. It is important to note that a part of a passage could be completely off-topic, while another one is relevant; this is the case because some chunks are relatively big and contain a lot of diverse information or show different intuitions in solving exercises
1 — Marginal	Same chapter/theme, but no part clearly helps; or only loosely related. The floor for "right area, wrong content."
0 — Not relevant	Different chapter/theme, and nothing in the passage helps (e.g. a probability passage for an exponential/logarithm question).

A same-chapter passage scores at least 1 (0 is reserved for a different chapter contributing nothing); utility overrides topic upward (a part helping a part scores 2 even from another chapter); a required method/format/procedure counts (2, or 3 if it answers the whole question); and each passage is judged alone. The seven decision rules in full and the four-level worked example are in Annex A and Annex M.A.

M7.2.3 Scientific grounding: a domain adaptation of a standard scale

The scale uses the standard graded-relevance scale of the TREC Deep Learning tracks, whose four NIST assessor levels are reproduced verbatim by the Bing/UMBRELA LLM relevance-assessor (Thomas et al. 2024; Upadhyay et al. 2024), mapping almost one-to-one:

This rubric	TREC DL / Bing / UMBRELA
3 — sufficient (answers all parts)	3 — perfectly relevant: "dedicated to the query and contains the exact answer"
2 — a part helps a part	2 — highly relevant: "has some answer… may be unclear, or hidden amongst extraneous information"
1 — same chapter, does not help	1 — related: "related to the query but does not answer it"
0 — different chapter, no help	0 — irrelevant: "the passage has nothing to do with the query"

Grade 1 ("right topic, but does not help") may look unusual, but it is TREC's standard "related" level. The rubric departs from TREC in two deliberate ways: - "same topic" means "same syllabus chapter", which is what separates 0 from 1 - A passage is relevant if a correct solution would use it (§M7.2). The question is "would a correct solution use this passage?", not TREC's "does it contain the answer?".

The chapter-level rule only decides between 0 and 1. Since Recall@k counts only grades ≥ 2 (§M7.2.4), it leaves Recall@k unchanged; in nDCG@k it changes a passage's gain from 0 point to 1 point, small next to 3 points for a grade 2 and 7 points for a grade 3.

M7.2.4 From graded labels to metric relevance

Both rules were fixed before any results, so the threshold could not be chosen after the fact. Recall@k treats a passage as relevant only if it is graded 2 or 3; the 1↔2 boundary is therefore the relevance threshold, which is why the pilot focused on it (§M7.3). nDCG@k uses the grade itself: each passage earns 2^grade − 1 points (0, 1, 3 or 7), passages lower in the list count less (divided by $\log_2(\text{rank} + 1)$), and the total is divided by the best score possible for that query, giving a value between 0 and 1. Both are implemented in src/eval/relevance/retrieval_metrics.py.

M7.3 Calibration pilot of the rubric

Before the full dispatch, a pilot tested the rubric: two tutors independently graded 80 pairs (5 stratified queries). Agreement was measured with weighted Cohen's κ (more info in §M9), a two-rater agreement score corrected for chance. On the full 0–3 scale, $\kappa = 0.704$, and 96.2% of pairs were within one grade of each other. But when grades are reduced to relevant (≥ 2) or not, κ fell to 0.587 — the weak point, since ≥ 2 is the Recall threshold. Of the 32 pairs the tutors graded differently, 13 were split between 1 and 2; disagreements concentrated on property-heavy queries (e.g. a BAC integral-parity item).

Based on that pilot's result, the response was to reframe the guiding question to "would a

correct solution use this?", adding the "je m'en sers" check (Annex A), sharpening the 0↔1 and 2↔3 boundaries, and adding a four-level worked example.

The final version writes down the oral clarifications given at dispatch, without changing the definition. It is the version the LLM assessor is prompted with (§M8), so machine and panel judge the same applied definition. Full changelog: Annex M.B and decisions/2026-06-18-rubric-v1.0.md.

M7.4 Design choices: The annotation plan

M7.4.1 The overlap design

Two demands pull against a fixed budget: coverage (every one of the 100 queries needs its pairs labelled at least once, or the qrels behind Recall@k / nDCG@k are incomplete) and agreement (an inter-rater statistic such as Krippendorff's α is only defined on items more than one annotator graded). The design splits the work into two tiers, one per demand. Assignment is by query, not by pair: each query's ≈ 15.7 passages go to a single grader, who reads the question once and then grades its passages one after another, each on its own (rule 6, Annex A). Pooling retrieved candidates and judging per-(query, chunk) relevance is the standard IR evaluation method, used to build qrels since the earliest TREC campaigns (Voorhees 2000) and carried into retrieval-for-RAG benchmarking (Upadhyay et al. 2025).

M7.4.2 The anchor and coverage tiers

- Anchor (agreement) tier: a stratified-random 10 queries ($\approx$ 157 pairs), each graded independently by 2 annotators. It yields both the human inter-annotator ceiling and, after the reframe, the judge–annotator calibration of the LLM assessor (§M8).
- Coverage tier: the remaining 90 queries ($\approx$ 1,408 pairs), each graded by one annotator. The H1/H2 metrics use all 100 queries, not just the 90 coverage ones. Anchor pairs enter the qrels as the average of their two grades, rounded half up.

Two raters per anchor pair is the minimum at which an agreement statistic is defined (Artstein and Poesio 2008), a redundancy the pilot validates by reaching a usable $\kappa = 0.70$ at that level.

The anchor is stratified-random, not hand-picked — its 10 queries mirror the 60/20/20 composition of the full set (§M3): 6 single-leçon, 2 multi-notion, 2 hard-edge. At 10 queries it spans only 7 of the 10 chapters, an accepted trade-off of the load reduction (§M7.4.3) and a limitation on the generality of the ceiling figure.

M7.4.3 Load balancing of the query pairs and the design revision

The binding constraint is per-annotator load. In fact, a heavier 20-query × 5-rater first cut placed ~283 pairs on each annotator. Because of the raters' fatigue, it was rebalanced to the 10-query × 2-rater anchor plus 90 coverage queries (~172 pairs each, a ~39% reduction), accepting lower anchor redundancy and the narrower chapter span. This is justified by the pilot's $\kappa = 0.70$ at redundancy 2. How queries were split between raters (the load arithmetic, the ±7% greedy allocation, the sliding-window rater pairing, and the seed-42 generation) is in Annex M.C.

M7.4.4 Dispatch, versioning, the panel, and bias control

The full **dispatch** covered 1,565 unique (query, chunk) pairs (`results/relevance_pairs.jsonl`), each stamped with its `rubric_version` in case the rubric is changed later; counting the anchor tier's double-grading, this is 1,721 total 0–3 judgements to be made. The **rubric is frozen after dispatch** (any later change requires a version bump and re-annotation) and annotator identities are anonymised.

Crucially, the **panel** is not ten teachers but mixed and curriculum-proficient: two mathematics tutors, six former Terminale C students, and two current Terminale C students. They were selected because the relevance task needs recognising whether a correct Terminale C solution would use a passage, i.e. recent mastery of exactly this curriculum. The consequence is that the coverage qrels are panel-produced, predominantly by students, recorded as a limitation in §M9.

Regarding bias control, two ordering choices protect the agreement machinery and, through it, H3. First, no LLM pre-labelling of the human pass: annotators never see a machine-suggested grade before judging, which would anchor them and inflate apparent judge–annotator agreement; the assessor labels pairs only after the clean human gold is fixed, and that ordering is the H3 protocol (§M8). Second, the response-grading track is out of scope here: this design covers (query, chunk) relevance only; the tutor-answer multi-criterion grading is a separate track that was not dispatched (§M10). Both keep the agreement statistic a measure of human judgement uncontaminated by model output.

M8. The LLM relevance-assessor (H3)

M8.1 The reframed hypothesis and why it is executable

The original H3 asked whether an LLM could grade complete generated responses on the four-criterion rubric in agreement with experts (§2.4). That requires a (query, response) gold set, which could not be produced on time. Only the (query, chunk) relevance package was dispatched and completed. With the deadline close it was reframed: the annotation object was changed but not the ambition; the LLM now produces relevance labels on (query, chunk) pairs, the same object the human panel already labelled. Hence, the gold pairs exist, the agreement machinery already runs on this object, and no new human labour is required:

> **H3 (reframed)**. A calibrated LLM relevance-assessor, scoring (query, chunk) pairs under the same rubric the human panel used, (i) gives the same H1 and H2-retrieval verdicts as the human qrels when the four retrievers are scored against its labels (ii) reaches per-item agreement with held-out human annotators that approaches the human inter-annotator ceiling.

The method is anchored to the information retrieval evaluation literature on LLM-as-relevance-assessor (Faggioli et al. 2023; Thomas et al. 2024; Upadhyay et al. 2024; §2.3): the thesis does not propose it but contributes its domain-transfer validation to a francophone, low-resource, curriculum-mathematics setting with a curriculum-proficient panel rather than crowd labels.

M8.2 The assessor: inputs, prompt, and decoding

The assessor is Claude Opus 4.8 (`claude-opus-4-8`). It is given each pair and asked for the same 0–3 score on the same rubric the panel applied (§M7.2). Following the finding that a formalised statement of the information needed improves human–LLM agreement (Thomas et al. 2024), it receives the rubric itself (the guiding question, the "would a correct solution use this?", and the decision rules) as its instruction, so human and machine judge against the same model-independent definition. Three controls are fixed and reported with the run:

- **Decoding fixed, determinism tested**. The model exposes no `temperature`, so reproducibility is established by test–retest reliability rather than assumed: each item run twice and aggregated by the round-half-up mean (the self-consistency idea of Wang et al. 2023). Two runs is a deliberately coarse check, but stricter than the single-run protocol common in the literature; the measured reliability is reported with the results.
- **The prompt is locked** before the assessor sees any pair, and paraphrase sensitivity is recorded as a limitation (Thomas et al. report LLM relevance judgements can shift under re-wording). The locked system prompt (sha256 `prompt_id = ba9d533a88b87292`, recorded in every run's `run_meta.json`) is reproduced verbatim in Annex B.
- No human pre-labelling precedes the assessor, and it never anchors the humans: the human gold is fixed first, then the assessor labels (§M7.4.4). This ordering is the H3 protocol.

M8.3 Success criterion: the human ceiling, not a fixed threshold

A flat "Krippendorff's $\alpha \geq 0.7$" bar was initially set, but later updated: the claim to defend is that assessor–human agreement lands inside the human-human range. In other words, the assessor agrees with held-out annotators about as well as they agree with one another following the metrics defined (Krippendorff's α, Cohen's κ). This is the honest yardstick because the literature consistently reports human–LLM agreement below human–human (Faggioli et al. 2023; Thomas et al. 2024). Because the panel is mixed, this ceiling is a mixed-expertise one and the assessor "approaching" it must be read in that light.

M8.4 The two-step validation of the assessor

Primary Validation. The headline test is the field's primary validation mode (Thomas et al.; UMBRELA; §2.3) and the one aligned with what the assessor is for. It is a system-ranking agreement instead of a per-item agreement, which is stronger and more relevant than a generic agreement coefficient. (1) run the assessor over the (query, chunk) set to produce LLM qrels (two runs aggregated, no humans); (2) recompute Recall@5 and nDCG@5 for all four retrievers on the LLM qrels, reusing the existing metric code; (3) compare the retriever ordering under the LLM qrels to that under the human qrels (exact ordering agreement, pairwise concordance, and the Kendall's τ meausre of sytem ranking similarity). If the LLM qrels reproduce the H1 (hybrid > dense) and H2-retrieval verdicts, the assessor is demonstrably good enough for the exact decision it supports. *Caveat*: with only four systems an ordering statistic is coarse, so τ is read with care. The two orderings and their agreement are reported with the results.

Secondary validation. Here comes the per-item agreement against the ceiling. This finer view places judge–human per-item agreement against the human–human agreement on the anchor

overlap (§M7.4.2): Krippendorff's α for assessor-vs-panel against panel-vs-panel α on the same items. It is secondary (a sub-ceiling α is expected, not disqualifying as mentioned in §M8.3) and serves to locate where the assessor diverges, broken down by the contested boundaries (the 1↔2 threshold especially).

M8.5 Circularity: the concern and why the design contains it

The literature raises a specific circularity risk (Faggioli et al. 2023; Thomas et al. 2024): LLM-produced relevance labels may be biased toward retrievers that themselves rank by LLM-like semantic similarity. For this case here, a tilt toward the dense channel over lexical BM25 can be expected. Hence, the design contains this by making the verdicts of the H1 and H2 hypotheses on the human-panel qrels, never on the LLM qrels. The tilt remains a limitation of the assessor itself: if its labels replaced human ones to compare retrievers, those that rank by semantic similarity could be favoured.

M8.6 Scope of the claim

The reframed H3 validates an evaluator: whether an LLM can stand in for human relevance labels well enough to rank retrievers as humans do. It does not show the tutor generates better answers — that is the work of the response judge (original H3, now H3b; harness built and unit-tested at `future_work/response_judge_h3b/`, but unrunnable without (query, response) gold; §M10), the H2 generation side, and the H4 comparison, all carried as future work. Together with H1 and the H2 retrieval side, the reframed H3 establishes the retrieval-and-evaluation foundation on which that generation-quality study is the immediate next step.

M9. All metrics and threats to validity

M9.1 Statistical methods

Retrieval metrics. Each query is scored against the human qrels at cutoffs $k \in \{5, 10\}$. A chunk counts as relevant when its grade is ≥ 2, and a chunk that no annotator judged counts as 0. The pedagogical retriever returns exactly five chunks, so H2 is compared at $k = 5$ only. Queries with no chunks graded 2 or higher have recall and nDCG undefined and are left out of the averages for those two metrics.

- **Recall@k**: the share of the query's relevant chunks that appear in the top k.
- **nDCG@k** (normalised discounted cumulative gain): uses the full 0–3 grade. Each retrieved chunk earns a gain of $2^g - 1$ (0, 1, 3 or 7 points), divided by $\log_2(\text{rank} + 1)$ so that lower ranks count less. The total is divided by the total of the best possible ordering of the query's judged chunks, which gives a score between 0 and 1.
- **MRR** (mean reciprocal rank): 1/rank of the first relevant chunk (0 if none is retrieved), averaged over queries.

Agreement coefficients. Both compare the agreement observed with the agreement expected by chance: the highest is 1 = perfect agreement.

- **Krippendorff's** α (ordinal): $\alpha = 1 - D_o/D_e$, the observed disagreement over the disagreement expected by chance. It accepts any number of raters and missing grades, which

suits the anchor design (ten raters, a different pair of raters on each query). Because it is ordinal, a 0 vs 3 disagreement weighs more than a 1 vs 2. Used for the human ceiling on the 155 double-rated anchor pairs, for the assessor against the humans, and for the assessor's test–retest reliability across its two runs.

- **Cohen's** κ: for exactly two raters, $\kappa = (p_o - p_e)/(1 - p_e)$, with p_o the observed and p_e the chance agreement. Reported in two forms: quadratic-weighted on the 0–3 scale (a disagreement of d grades weighs $d^2/9$), and unweighted on the binary relevant/non-relevant split at grade 2.

Significance and robustness (H1 and the H2 retrieval side).

- **Paired query-level bootstrap** (B = 2,000, seed 42): (Efron & Tibshirani 1993; for IR metrics, Sakai 2006) the 100 queries are resampled with replacement, both retrievers are scored on the same resampled set in each replicate, and the per-replicate difference yields a 95% percentile confidence interval and the share of replicates in which the difference is positive. This is a resampling interval for a fixed query set, not a parametric p-value.
- **Label-perturbation sensitivity**: the H1 (k = 5, k = 10) and H2 (k = 5) differences are recomputed on fifteen perturbed versions of the labels: a relevance cut at 1 or 3 instead of 2; severity recalibration, where each rater's grades are shifted so their mean matches the panel's; anchor pairs resolved by the higher or the lower of the two grades instead of their rounded mean; and each of the ten annotators left out in turn.

Ranking agreement (H3 primary test). The four retrievers are ordered by nDCG@5 under the LLM qrels and under the human qrels, and the two orderings are compared by:

- **Exact ordering agreement**: whether all four retrievers come out in the same order.
- **Pairwise concordance**: the share of the six retriever pairs that both rankings put in the same order.
- **Kendall's** τ**:** $\tau = (C - D)/6$, with C the concordant and D the discordant pairs. It runs from -1 (reversed) to $+1$ (identical). It is coarse with only four systems, because a single swap moves it by 0.33.

Table M1. Summary of the measures in §M9.1. g_i = grade of the chunk at rank i; N = number of queries; D_o, D_e = observed and chance-expected disagreement; o_{ij}, e_{ij} = observed and chance-expected counts of the grade pair (i, j); p_o, p_e = observed and chance agreement; C, D = concordant and discordant retriever pairs.

Measure	Formula	In brief
Recall@k	$\dfrac{\text{relevant chunks in top } k}{\text{relevant chunks for the query}}$	Share of the relevant chunks (grade ≥ 2) that are retrieved. H1, H2.
nDCG@k	$\dfrac{\text{DCG@}k}{\text{ideal DCG@}k}$, $\text{DCG@}k = \sum_{i=1}^{k} \dfrac{2^{g_i} - 1}{\log_2(i+1)}$	Graded quality of the top k, rewarding high grades near the top; 0 to 1. H1, H2, H3 ranking.
MRR	$\dfrac{1}{N} \sum_q \dfrac{1}{\text{rank of first relevant chunk}}$	How early the first relevant chunk appears. H1, H2.

Measure	Formula	In brief
Krippendorff's α (ordinal)	$1 - D_o/D_e$	Agreement beyond chance; any number of raters, missing grades allowed. Human ceiling, assessor vs humans, test–retest.
Cohen's κ, quadratic-weighted	$1 - \frac{\sum_{ij} w_{ij}\, o_{ij}}{\sum_{ij} w_{ij}\, e_{ij}}$, $w_{ij} = (i-j)^2/9$	Two-rater agreement on the 0–3 scale; distant grades weigh more. Pilot, two-rater summary.
Cohen's κ, binary	$\frac{p_o - p_e}{1 - p_e}$	Two-rater agreement on relevant vs not relevant (grade ≥ 2). Assessor vs humans at the 1↔2 boundary.
Exact ordering agreement	1 if both orderings are identical, 0 otherwise	Do LLM and human qrels rank the four retrievers the same way? H3.
Pairwise concordance	$C/6$	Share of the six retriever pairs ordered alike. H3.
Kendall's τ	$(C - D)/6$	Ranking similarity, from -1 (reversed) to $+1$ (identical). H3.

M9.2 Threats to validity

- OCR quality: the OCR error rate (prose CER 11.51%, equation exact-match 87.46%; Annex E) was measured on a single hand-transcribed chapter using marker's plain mode, while the indexed corpus was produced with its LLM/vision mode. The quality of the indexed text is therefore assumed from that chapter, not measured. Every retrieval result rests on this text.
- Gold provenance: the coverage qrels (90% of all qrels) are single-rater, and 8 of the 10 raters are former or current Terminale C students rather than teachers. Where two raters graded the same pair, they agreed only moderately (ordinal $\alpha = 0.46$ on the 155 anchor pairs, §4.1.1). The label-perturbation analysis (§M9.1) shows that the H1 and H2 verdicts hold in all 16 label scenarios
- Assessor circularity and paraphrase sensitivity: LLM labels may tilt toward the dense channel, contained because the retrieval verdicts conclude from human qrels, never LLM qrels; the single locked prompt makes paraphrase sensitivity a recorded limitation.
- Contamination and leakage: BAC 2025 is held out to limit leakage, but part of the measured H1 effect may be attributable to lexical overlap from the synthesised-query construction, and the held-out slice (n = 9) is too small to settle external validity. Worked-example contamination is $\approx$ 32% of the `theory`-labelled chunks (95% CI 20–48%, n = 40), about 12% of the theory bucket; a full-corpus estimate is future work.
- Corpus and scope: figure-dropping (15.7% of chunks) and geometry exclusion bind every claim to non-geometry, text-only Terminale C content.
- Statistical power: 100 queries, four retrievers, and a 155-pair anchor overlap limit statistical power and generality.

M10. Generation-side and student evaluation (specified, not executed)

Two further evaluations were designed but not conducted by the time of the report, because both depend on a (query, response) gold set.

(i) A response judge (the original H3, reframed to H3b) scores a full tutor answer on four model-independent criteria (factual accuracy, curriculum alignment, pedagogical quality, language register), variant-blind and based on a correct human solution; its harness is built and unit-tested but unrunnable without response-gold and a finalised rubric. It is the instrument that would test the H2 generation-side claim: whether the theory-plus-example mix yields better answers, not just better retrieval (which §M5.4 stresses cannot be concluded from retrieval metrics). Used to score frontier and small-model answers, each with and without retrieval, it would also test H4.

(ii) A small field study with 10–20 Terminale C students (fixed problem set scored for correctness; pre/post confidence; blind pairwise A/B; short interview) would provide student-side H2 evidence; nothing is built and it needs a cohort and ethics turnaround unavailable before the deadline. Both are carried as future work and the full protocols are in Annex M (M.D).

Chapter 4. Results

This chapter presents the empirical results for the three hypotheses executed within the time and data available: H1, the H2 retrieval side, and the reframed H3. It is organised so the evidence is read in the order in which it is trusted: first the human inter-annotator agreement that bounds how far the gold labels can be relied upon (§4.1), then the two retrieval comparisons those labels adjudicate (§4.2, §4.3), then a consolidated view of all four retrievers (§4.4), then the validation of the LLM relevance-assessor the same labels make possible (§4.5), and finally what is not a result here (§4.6).

Two reading conventions apply throughout. First, retrieval differences are reported as a point estimate Δ with its 95% paired-bootstrap CI in brackets and the bootstrap probability $P(\Delta>0)$ that the difference carries its hypothesised sign; an interval excluding zero is read as unlikely to be an artefact of the fixed 100-query set, and every verdict is additionally stress-tested under the label-perturbation analysis of §4.4.2. Second, the H1 and H2 retrieval verdicts are computed against the human-panel qrels, never the LLM qrels — the LLM qrels exist only to test the assessor in §4.5.

4.1 Reliability of the gold labels

The retrieval comparisons are only as trustworthy as the relevance labels they are scored against, and the reframed-H3 ceiling of §4.5 is that human agreement. This section reports the inter-annotator agreement that bounds how far the labels can be trusted (§4.1.1), then the dataset profile and quality-control checks on the labels themselves (§4.1.2).

4.1.1 Inter-annotator agreement

Agreement is computed over the anchor overlap of the disp10 design — the subset of (query, chunk) pairs labelled by more than one annotator. Two properties of the panel bear on how the number is read: it is a mixed-expertise ceiling (agreement across the two teachers and eight students, not a pure-expert standard), and the coverage qrels behind H1/H2 are mostly student-produced single-rater labels.

Table 1. Human inter-annotator agreement on the disp10 anchor overlap. Ordinal Krippendorff's α (0–3 scale) is the primary statistic; nominal α and the two-rater summary are reported alongside.

Statistic	Value
Krippendorff's α — ordinal (0–3)	0.46
Krippendorff's α — nominal	0.16
Units (total / multi-rated)	1,565 / 155
Annotators	10
Two-rater exact agreement	38.7%
Two-rater adjacent agreement (± 1)	87.7%
Two-rater weighted κ	0.47

The ordinal α is the headline because relevance is an ordered 0–3 judgement: an off-by-one disagreement (a 2 vs. a 3) is less severe than a polar one (a 0 vs. a 3), and the ordinal weighting encodes exactly that. The ordinal α of 0.46 is a moderate mixed-panel ceiling, and the comparison of exact (38.7%) to adjacent (87.7%) agreement shows why it is not higher: only one co-rating in eight (12.3%) is a disagreement of more than one grade, so the bulk of the residual divergence is threshold noise around the soft 1↔2 cut — the boundary between "not used by a correct solution" and "used" — rather than substantive disagreement about whether a chunk is on-topic. The large gap between the ordinal α (0.46) and the nominal α (0.16) is the same fact from the other side: treated as unordered categories the labels look weak, but the disagreements are overwhelmingly near-misses on an ordered scale, which the nominal coefficient cannot see. This moderate-but-orderly agreement is what licenses reading the H1/H2 differences as point estimates against a noisy gold, and is exactly why each is additionally given a bootstrap interval.

By the Landis and Koch (1977) benchmark, 0.46 is "moderate" (0.41–0.60) — below the thresholds conventional for content analysis, but those presume a single true category, which graded relevance is not. The relevance-assessment literature treats moderate overlap as the expected operating condition: Voorhees (2000) found that substantially different relevance-judgment sets still induced highly stable system rankings, and Bailey et al. (2008) found that mixed expert/non-expert panels agree less yet remain usable for ranking systems. A two-teacher, eight-student panel is therefore expected to land at a moderate ceiling — the leave-one-out spread of 0.34–0.59 (§4.4.2) confirms this is a property of the whole panel, not one weak rater — which is exactly why the H3 validation leads with ranking agreement and treats per-item α as secondary. Fuller interpretation is deferred to Chapter 5.

4.1.2 Annotation dataset properties and quality control

Because the retrieval verdicts are scored against these labels, the labels themselves are profiled and quality-controlled before any metric is read. Three properties matter.

Relevance base-rate and rater severity. Across all 1,720 labels the grade distribution is skewed low (0: 30.5%, 1: 32.3%, 2: 24.9%, 3: 12.3%; mean 1.19, 37.2% relevant at the ≥ 2 cut) — the direct cause of the low nominal α. More consequential is severity heterogeneity: per-annotator mean grade ranges from 0.80 to 1.69, a better-than-twofold spread in how readily a rater calls a chunk relevant. Two raters are markedly lenient (mean 1.69, 1.68), two markedly severe (0.80, 0.89), one defaults heavily to grade 1, and one uses the scale near-bimodally. This is the

heterogeneity the sensitivity analysis (§4.4.2) is designed to absorb, reported here rather than hidden.

Relevance by content type — bearing on H2. Relevance is essentially flat across chunk content types: worked examples (mean 1.20) and solutions (1.18) are judged about as relevant as theory (1.24) or theorems (1.27) — lower by ≤ 0.09 on the 0–3 scale, and no lower on the binary ≥ 2 cut Recall uses (solutions are judged relevant 41.0% of the time, against theory's 40.3%). This matters for §4.3: the pedagogical retriever's enforced example slots are not seating systematically lower-relevance material, so any retrieval-side H2 deficit cannot be explained by examples being judged irrelevant — it must come from displacing higher-ranked relevant chunks, a near-relevance-neutral substitution. Consistently, the labelled pool precision of the pedagogical retriever's returned chunks is the highest of the four variants (47.6% relevant, vs. 43.8% hybrid, 39.5% BM25, 39.2% dense).

Coverage and integrity. The judged set covers all 100 queries with a median of 5 relevant chunks per query (4 queries have no chunk judged relevant at ≥ 2 and are NaN-guarded out of the per-query means). A structural quality-control pass found no integrity faults — no duplicate pairs, no labels outside the dispatched set, no query/chunk-id inconsistency, no out-of-range grades — and no evidence of end-to-end file misalignment: each rater's mean absolute disagreement with its anchor partners is within the panel norm (largest deviation $z = +0.55$ against a flag threshold of 1.5), and grade-entropy and longest-run checks show no straight-lining. A single localised likely-error (one severe rater inverted its grades on one differential-equation query) was left uncorrected, because post-hoc editing of labels to raise agreement would compromise the gold, and §4.4.2 shows no single label or rater drives the verdicts. The labels are sound and are used in full.

4.2 H1 — hybrid vs. dense retrieval

H1 is the hypothesis that the RRF hybrid outperforms dense-only retrieval on the 100-query set, scored against the human-panel qrels v1.0. The registered primary metric is nDCG@5; Recall@5 and MRR are reported alongside, with the $k = 10$ comparison in §4.4.

Table 2. H1: hybrid (RRF) vs. dense-only on the 100-query set, human-panel qrels v1.0. The last two rows split nDCG@5 by query source (synthesised / held-out BAC-2025).

Metric	Dense	Hybrid (RRF)	Δ (Hyb − Dense)	P(Δ>0)
Recall@5	0.373	0.411	+0.038 [−0.009, +0.085]	0.95
nDCG@5 (primary)	0.476	0.525	+0.049 [+0.009, +0.093]	0.99
MRR	0.612	0.663	+0.051 [−0.012, +0.115]	0.95
nDCG@5 (synth, n=91)	0.476	0.540	+0.064	—
nDCG@5 (BAC25, n=9)	0.472	0.381	−0.091 [−0.315, +0.097]	0.21

Three secondary analyses accompany the main contrast. Leakage control: the last two rows

split nDCG@5 over the 91 synthesised queries and the 9 held-out BAC-2025 queries; BAC 2025 is de-indexed from the corpus, so a hybrid advantage surviving on that slice — where no source passage can leak — would be evidence the H1 effect is not an artefact of train/test overlap (the n = 9 slice is read as corroboration, not an independent test). Query-type stratification: the per-bucket hybrid advantage in nDCG@5 is single-leçon +0.065 (n = 60), multi-notion +0.040 (n = 20), hard-edge +0.015 (n = 20) — largest on the simplest single-topic queries and smallest on the hard-edge ones, the opposite of the hypothesis that the sparse channel earns its place on rare-term matches. Three-way channel comparison: the hybrid's nDCG@5 (0.525) exceeds both BM25 (0.489) and dense (0.476) alone, so its gain is a genuine fusion benefit, not merely the better of the two channels (full grid in §4.4).

Interpretation. H1 is supported, with one honest qualification about why it holds. On the primary metric the hybrid beats dense by +0.049 nDCG@5, with a 95% bootstrap CI of [+0.009, +0.093] that excludes zero; it is a genuine two-channel fusion, and it is robust to every annotator scenario of §4.4.2 and to dropping any single rater. The qualification concerns the effect's source. The advantage is carried by the sparse (BM25) channel, which behaves very differently across query sources: on the 91 synthesised queries BM25 alone (0.498) outperforms dense (0.476), whereas on the 9 held-out BAC-2025 queries BM25 (0.390) underperforms dense (0.472). The synthesised queries also share more vocabulary with their relevant chunks than the real exam queries do (55.5% vs. 45.6% query-term coverage), and the hybrid gain concentrates on precisely the queries where that overlap is highest. Part of the measured H1 advantage is therefore attributable to lexical overlap between the synthesised evaluation queries and the indexed corpus — a property of how the evaluation set was built, not necessarily of how a student would phrase a question. This is not read as a held-out reversal: the −0.091 on the BAC-2025 slice comes from two of nine queries (one with a single relevant chunk, making its nDCG near-degenerate), and its CI [−0.315, +0.097] is too wide to be informative — nine queries cannot settle the question. The honest verdict: H1 holds on this corpus and is statistically supported on the primary metric, but leans on lexical matching that the synthetic-query construction advantages; confirming the hybrid gain on real student queries is the natural next test. The dense (semantic) channel is the one that holds up on the real held-out questions and is the more conservative choice where query phrasing cannot be assumed to track the corpus.

4.3 H2 (retrieval side) — pedagogical vs. hybrid retrieval

It must be stressed that nDCG@5 is neither necessary nor sufficient for the H2 pedagogical-quality claim. The pedagogical retriever deliberately passes over higher-relevance theory chunks to seat its two example slots, so it can lose on nDCG@5 and still tutor better, or win with no tutoring gain. The retrieval-side contrast therefore tests one thing only: whether the enforced theory–example mix is achieved without sacrificing relevance. The pedagogical-quality verdict itself is not delivered in this thesis — it depends on the generation-side evaluation, which is future work (§4.6).

Table 3. H2: pedagogical vs. hybrid at the matched cut-off k = 5. The "cond." row restricts to the 98 queries whose fused pool held ≥ 1 example-type chunk (no example-slot fallback).

Metric	Hybrid	Pedagogical	Δ (Ped − Hyb)	P(Δ>0)
Recall@5	0.411	0.377	−0.034 [−0.061, −0.007]	0.004
nDCG@5	0.525	0.504	−0.022 [−0.043, +0.001]	0.03
MRR	0.663	0.627	−0.035 [−0.080, +0.006]	0.05
nDCG@5 (cond.)	≈0.525	≈0.503	−0.022 (n = 98)	—

Two features shape how the numbers read. The marginal vs. conditional split: on example-poor queries the fallback rule backfills the example slots with theory content, degenerating the pedagogical retriever toward the hybrid baseline. The marginal row (all queries) is the realistic operational estimate; the conditional row (no fallback) is the cleaner test of the principle. Here the fallback fires on only 2 of 100 queries (example-availability 98%; mean 1.96 of 2 example slots filled), so the two estimates are effectively identical (both −0.022 nDCG@5). And a conservative bias from baseline contamination: a manual audit of `theory`-labelled chunks (n = 40, seed 42) estimates that ≈ 32% are in fact headingless worked examples (95% CI 20–48%), about 12% of the theory bucket, so the hybrid baseline can already surface example-like content through its theory channel and partially close the gap the pedagogical retriever enforces by construction — meaning a positive H2 effect would appear in spite of a baseline that captures some of the same intent.

Interpretation. The H2 necessary-condition is not met: the pedagogical retriever is, by a small but consistent margin, below hybrid on relevance. At the matched k = 5 it trails hybrid by −0.034 Recall@5 (CI [−0.061, −0.007], excluding zero) and −0.022 nDCG@5; the result is stable across the sensitivity scenarios (Recall@5 Δ in [−0.041, −0.026] throughout) and barely moves under example-availability conditioning. Three facts keep this from being a refutation. First, the deficit is small and arises despite the conservative baseline contamination above. Second, example and solution chunks are judged as relevant as theory, and the pedagogical retriever's returned set has the highest labelled pool precision of the four variants (47.6%); it is not seating low-relevance material, it is making a near-relevance-neutral substitution that trades a little ranking relevance for a guaranteed theory-plus-example structure. Third, and decisively, nDCG@5 is the wrong instrument for the H2 claim: per-chunk relevance cannot reward the pairing of a theory chunk with its worked example, which is the entire pedagogical hypothesis. The retrieval-side result therefore establishes only that the enforced mix is achieved at a small, bounded relevance cost — a necessary-conditions check, passed in the weak sense that the cost is small — and emphatically not the pedagogical-quality verdict, which depends on the generation-side evaluation this thesis does not execute. (The −0.30 figure a naïve k = 10 comparison would produce is an artefact of scoring a fixed 5-chunk retriever against a 10-chunk one, and is not reported as a result.)

4.4 Consolidated retrieval results (all four variants)

For completeness, and to support the H1 three-way analysis and the H3 ranking validation, the full metric grid for all four retrievers is reported in one place, then stress-tested against annotator behaviour (§4.4.2).

4.4.1 The four-variant comparison

Table 4. All four retrievers on the 100-query set, human-panel qrels v1.0. Best value per column in bold.

Variant	Recall@5	nDCG@5	Recall@10	nDCG@10	MRR
BM25	0.370	0.489	0.599	0.561	0.657
Dense	0.373	0.476	0.590	0.552	0.612
Hybrid (RRF)	0.411	0.525	0.678	0.613	0.663
Pedagogical	0.377	0.504	—†	—†	0.627

† The pedagogical retriever returns a fixed 5-chunk bundle, so its top-10 equals its top-5 and the k = 10 cells are undefined; H2 (§4.3) is therefore scored only at the matched k = 5.

The ranking the four variants induce is itself a result: it is the human-qrels ordering against which the LLM assessor's ordering is compared in §4.5. By the primary nDCG@5 the order is hybrid (0.525) > pedagogical (0.504) > BM25 (0.489) > dense (0.476); by Recall@5 it is hybrid (0.411) > pedagogical (0.377) > dense (0.373) > BM25 (0.370). In both, hybrid is first and dense/BM25 are last, with the pedagogical retriever a close second — a consequence of its high pool precision offsetting its lost ranking relevance.

4.4.2 Robustness of the retrieval verdicts to annotator behaviour

Because the H1 and H2 verdicts are read off labels from a severity-heterogeneous panel, each verdict is re-derived under a menu of plausible annotator-behaviour scenarios. The question is not whether the point estimates move — they do, slightly — but whether the direction of each verdict survives.

Table 5. Retrieval verdicts under qrels-perturbation scenarios (H1 = hybrid − dense, nDCG@10; H2 = pedagogical − hybrid, nDCG@5). "Recalibrated" mean-centres each rater; "optimistic/pessimistic" resolve each multi-rated pair by max/min instead of the mean; leave-one-out drops each annotator in turn.

Scenario	α	H1 (nDCG@10)	H2 (nDCG@5)
Baseline (mean, t $\geq$ 2)	0.46	+0.061	−0.022
Threshold t $\geq$ 1	0.46	+0.061	−0.022
Threshold t $\geq$ 3	0.46	+0.061	−0.022
Recalibrated (severity removed)	0.46	+0.061	−0.022
Optimistic (max per pair)	0.46	+0.064	−0.022
Pessimistic (min per pair)	0.46	+0.060	−0.022
Leave-one-annotator-out (×10)	0.34–0.59	+0.056 … +0.070	−0.026 … −0.015

Across all sixteen scenarios the H1 advantage stays positive (nDCG@10 in [+0.056, +0.070]) and the H2 deficit stays negative (nDCG@5 in [−0.026, −0.015]); neither verdict flips, and

neither is driven by any single annotator. The leave-one-out column doubles as a diagnostic of the agreement: dropping the grade-1-defaulting or lenient raters raises α (to 0.55–0.59) while dropping the panel-consistent raters lowers it (to 0.34–0.40), and α never exceeds 0.59 — confirming the moderate ceiling is a property of a genuinely mixed-expertise panel, not one poor rater. With the bootstrap intervals of §4.2–§4.3 excluding zero, both directions are unlikely to be sampling noise. The one caveat the sensitivity analysis does not dissolve is the lexical-overlap question of §4.2, which concerns the construction of the evaluation queries rather than their annotation.

4.5 H3 (reframed) — the LLM relevance-assessor

The reframed H3 asks whether a calibrated LLM relevance-assessor, scoring (query, chunk) pairs under the same rubric the human panel used, can stand in for human labels well enough (i) to rank the four retrievers as the human qrels do and (ii) to reach per-item agreement approaching the human ceiling of §4.1. The success criterion is the ceiling, not a fixed 0.7 bar: the claim to defend is that judge–human agreement lands inside the human range.

The assessor is built and run: it scores every (query, chunk) pair on the 0–3 rubric with Claude Opus 4.8 under the locked prompt (reproduced in Annex B), grading all 1,565 pairs twice (3,130 judgments, zero errors) to produce the LLM qrels used below. Because the judge model exposes no temperature parameter, reproducibility is established as test–retest across the two runs.

4.5.1 Primary validation — does the assessor reproduce the retriever ranking?

The headline H3 test is system-ranking agreement, the field's primary validation mode (Thomas et al. 2024; UMBRELA): run the assessor to produce LLM qrels, recompute the metrics for all four retrievers against them, and compare the ordering to the human-qrels ordering of §4.4.

Table 6. Retriever ranking under human qrels vs. LLM-assessor qrels, by nDCG@5 (rank 1 = best; score in parentheses). The last rows give pairwise concordance over the six system pairs and Kendall's τ (coarse — only four systems).

Variant	Rank (human qrels)	Rank (LLM qrels)
Hybrid (RRF)	1 (0.525)	1 (0.571)
Pedagogical	2 (0.504)	2 (0.557)
BM25	3 (0.489)	4 (0.503)
Dense	4 (0.476)	3 (0.524)
Pairwise concordance	—	5 / 6
Kendall's τ (coarse)	—	+0.67

The decisive question is narrower than the table looks: does the LLM qrels reproduce the H1 verdict (hybrid > dense) and the H2-retrieval verdict (pedagogical vs. hybrid)? On both counts it does. H1 is reproduced: under the LLM qrels the hybrid still beats dense (+0.047 nDCG@5; +0.068 Recall@10 and +0.053 nDCG@10, the same direction and comparable magnitude as the human-qrels +0.061), so the assessor would have returned the same H1 decision. The

H2-retrieval verdict is reproduced: the pedagogical retriever again ranks second on nDCG@5 and does not exceed hybrid (−0.014 nDCG@5 under the LLM qrels, against −0.022 under the human qrels). Beyond the two registered verdicts, the full four-system ordering agrees on 5 of 6 pairwise comparisons (Kendall's $\tau = +0.67$); the single discordance is the BM25↔dense pair at the bottom, which the LLM qrels rank dense (0.524) just above BM25 (0.503) where the human qrels rank BM25 (0.489) just above dense (0.476). Two things make this swap minor. First, it is between the two weakest systems, on a gap of hundredths, leaving the top of the ranking — where the H1/H2 decisions live — untouched. Second, its direction is the recognised assessor-circularity risk made visible: the LLM qrels are uniformly more generous to the dense/semantic channel (dense nDCG@5 rises 0.476 → 0.524), the tilt recorded as a limitation of the assessor in §4.5.2 — never an input to the H1/H2 conclusions, which use the human qrels only. The headline is therefore that the calibrated assessor reproduces both decisions the human qrels are used to make — the strongest and most relevant form of the H3 ranking claim.

4.5.2 Secondary validation — per-item agreement against the ceiling

The finer view places the assessor's per-item agreement against the human ceiling of §4.1: Krippendorff's α (ordinal) for assessor-vs-panel labels, read against the panel-vs-panel α on the same anchor items. This is reported as secondary precisely because a sub-ceiling α is expected, not disqualifying — Faggioli et al. (2023) and Thomas et al. (2024) report LLM relevance judgments agreeing with human assessors considerably yet frequently below human–human levels (Cohen's $\kappa \approx 0.20$–0.64 across prompts, against ≈ 0.52 human–human).

Table 7. Assessor per-item agreement vs. the human ceiling: ordinal α and binary κ at the contested 1↔2 cut. "Like-for-like" scores the assessor against each raw human grade on the 155 anchor pairs by exactly the procedure that produces the human ceiling (no panel averaging).

Agreement	α (ordinal)	binary κ (1↔2)
Human ceiling — panel ↔ panel (155 anchor pairs)	0.46	0.26
Assessor ↔ individual human (155 anchor, like-for-like)	0.59	0.40
Assessor ↔ panel (all 1,565 pairs, collapsed)	0.55	0.42
Assessor test–retest reliability (run 1 ↔ run 2)	0.97	0.96

On the secondary criterion the reading is stronger than its "expected to fall short" framing anticipated: the assessor does not merely approach the human ceiling — it meets and exceeds it. On the like-for-like comparison, assessor↔human α is 0.59 against the 0.46 panel↔panel ceiling, and binary agreement at the 1↔2 cut is $\kappa = 0.40$ against 0.26; the collapsed all-pairs view corroborates at 0.55. This is not an averaging artefact — the like-for-like number, which never averages the humans, is the higher of the two. The mechanism is the one Voorhees (2000) anticipates: a single calibrated, near-deterministic rater (test–retest $\alpha = 0.97$) lands closer to each individual human, on average, than two independently noisy humans land to each other. It is read against a mixed-expertise ceiling: "exceeds the ceiling" means it exceeds the agreement this panel achieves, the relevant bar for whether the assessor can stand in for these labels. It

is also why the success criterion was set as the ceiling and not the literal H3 $\alpha \geq 0.70$ bar of Chapter 1: that bar is unreachable by the humans themselves on this mixed panel (0.46).

Two cautions are carried forward. First, the dense/semantic tilt is a recognised circularity risk: the assessor's per-item agreement is high partly because it, like the dense retriever, rewards semantic overlap, so a fraction of the agreement and the ranking reproduction may be shared bias rather than shared correctness — which is why the assessor's qrels never enter the H1/H2 verdicts. Second, the residual disagreement is overwhelmingly graded near-misses: 91% of assessor↔human anchor comparisons are within one grade, and the off-by-$\geq$2 cases ($\approx$ 9%) are where the assessor's limitation should be sought in future error analysis. The combined H3 verdict is therefore that the calibrated assessor reproduces both human-qrels retrieval decisions and agrees with individual teachers above the mixed-panel human ceiling, with a documented dense-channel tilt as its principal limitation — a domain-transfer validation of the LLM-as-relevance-assessor method for this low-resource francophone setting, not a claim that the judge is bias-free.

4.6 What is not a result here

To keep the contribution from being overread, the boundary of this chapter is stated explicitly. Three strands are executed and reported above — H1, the H2 retrieval side, and the reframed H3 — and all three are retrieval-and-evaluation results. None shows that the tutor generates better answers. The following are specified in the methodology and carried as future work.

- The response-grading judge (original H3 / H3b). A four-criterion judge (factual accuracy, curriculum alignment, pedagogical quality, language register; 1–5) over complete generated responses. Its harness is built and unit-tested but unrunnable for want of (query, response) gold, which was never collected. It is a different instrument on a different annotation object from the §4.5 assessor and produces no result here.
- The H2 generation side. Whether the enforced theory–example mix translates into measurably better tutoring responses — the pedagogical-quality verdict §4.3 does not deliver — depends on the response judge or a student A/B and is deferred.
- H4 (frontier vs. SLM under grounding) and the student study (correctness, confidence shift, A/B preference) are likewise unexecuted; both depended on the (query, response) gold and the human cohort that were out of reach for the submission.
- Generation-quality numbers of any kind. The generator is built, and its V2 prompt was selected by a 27-call controlled comparison on three single-extract items (§M6); that comparison checks the prompt's behaviour, not the quality of tutoring answers.

Read together, the executed strands establish the retrieval-and-evaluation foundation on which the generation-quality study is the immediate, fully-specified next step. The interpretation of that boundary — why a thesis of infrastructure results is defensible under the constraint — is taken up in Chapter 5.

Chapter 5. Discussion and Limitations

5.1 What the thesis establishes, and the honest exposure

The contribution of this thesis is a retrieval-and-evaluation foundation for a curriculum-grounded mathematics tutor in a francophone, low-resource setting: a chunked, content-typed Terminale C corpus, four retrieval variants with a registered H1 test and a pedagogical-retrieval H2 test on the retrieval side, a human relevance-annotation instrument with its agreement machinery, and a domain-transfer validation of the LLM-as-relevance-assessor method. These are real, executed contributions on data in hand.

The honest exposure — the one a juror is most likely to press, owned here rather than deflected — is that none of the three executed results demonstrates that the tutor produces better tutoring answers. H1 and the H2 retrieval side are about which passages are retrieved; the reframed H3 is about whether an LLM can label relevance like a human. All three are necessary infrastructure for a generation-quality study, and none is that study. This follows directly from the annotation object: a relevance label on a (query, chunk) pair says nothing about whether a generated response is pedagogically good, and the (query, response) gold that would license a response-quality claim was never collected.

The defensible framing — a methodological strength, not a save — is that this is what can be established rigorously under the constraint, and that it is the foundation the generation-quality study (§5.7) is the immediate next step from. The alternative — an under-powered, un-calibrated generation result on no gold data — would have been the less honest thesis. The H3 reframe is the clearest instance of this discipline: an unexecutable hypothesis was rebuilt onto data in hand rather than left to fail or faked.

5.2 Interpreting H1 (hybrid vs. dense)

The hybrid advantage holds (Table 2): +0.049 nDCG@5, with a 95% bootstrap CI of [+0.009, +0.093] that excludes zero, robust across every label-perturbation scenario. The natural reading is that lexical match contributes signal the dense channel misses — but the stratified analysis sharpens, and partly undercuts, it. The gain is not concentrated on the hard-edge queries (Δ +0.015), where a rare content-bearing token would be the deciding signal; it is largest on the simplest single-leçon queries (+0.065). And the leakage-control rows do not corroborate external validity: the sparse channel outperforms dense on the synthesised queries but underperforms it on the held-out real exam questions, which share less vocabulary with their relevant chunks. The most honest interpretation is therefore mixed: the hybrid advantage is real and statistically

supported on this evaluation set, but it is carried by lexical overlap that the synthetic-query construction inflates, so part of it may not transfer to how a real student phrases a question. The dense channel, which holds up on the held-out real queries, is the more conservative default. (The held-out slice itself, at n = 9 with a CI of [−0.315, +0.097], is too small to settle the question either way.)

The claim is bounded to this corpus, these untuned hyperparameters, and the non-geometry, text-only scope; confirming H1 on real student queries is the named next step.

5.3 Interpreting H2 (pedagogical retrieval, retrieval side)

The H2 retrieval-side result (Table 3) requires the most careful reading in the thesis, and the care is built into the design rather than added after the fact.

nDCG@5 is not the pedagogical-quality verdict. The pedagogical retriever deliberately passes over higher-relevance theory chunks to seat its two worked-example slots; it can therefore lose on nDCG@5 while assembling a better teaching context, or win with no teaching benefit. So the retrieval-side result tests one thing only — whether the enforced theory–example mix is achieved without sacrificing relevance — and the interpretation is modest in every branch: a small or null Δ means the mix is imposed at little relevance cost (the favourable outcome), while a material loss is read against the example-availability split, since a loss concentrated on example-poor queries (where the fallback degenerates the retriever toward the baseline) would be a corpus-coverage artefact rather than a refutation.

Observed: the favourable branch, essentially. The pedagogical retriever trails the hybrid by a small margin — −0.022 nDCG@5, −0.034 Recall@5 (the latter's CI excludes zero) — and the fallback fires on only 2 of 100 queries, so the marginal and conditional estimates coincide and the loss is not a coverage artefact. The mix is imposed at a small, genuine, but bounded relevance cost; combined with the finding that examples are judged as relevant as theory and that this retriever has the highest pool precision of the four, the retrieval side reads as near-relevance-neutral, with the pedagogical-quality question passing intact to the generation-side study (§5.7).

The conservative-bias caveat governs a null result. Because a pilot audit found some `theory`-tagged chunks are in fact headingless worked examples, the hybrid baseline can surface example-like content by chance, partially closing the gap the pedagogical retriever enforces. A null retrieval-side H2 result would therefore be ambiguous — no true effect, or a baseline that absorbed it — and the magnitude is not negligible: a manual audit (n = 40, seed 42) estimates ≈ 32% of the `theory`-labelled chunks are headingless worked examples (95% CI 20–48%), about 12% of the 514-chunk theory bucket. Resolving the ambiguity is exactly what the generation-side evaluation is for, which is why the pedagogical-quality claim is held pending (§5.7).

5.4 Interpreting the reframed H3 (the relevance-assessor)

H3 asks whether an LLM can stand in for human relevance labels well enough to (i) rank the four retrievers as humans do and (ii) agree per-item near the human ceiling. Two interpretive points hold regardless of the figures.

The primary read is the ranking, not the coefficient. If the LLM qrels reproduce the H1 and H2-retrieval verdicts (Table 6), the assessor is good enough for the exact decision it supports — a stronger claim than any single agreement number; a high per-item α with the wrong ranking would be less useful than a moderate α with the right one. On the executed run the LLM qrels reproduce both registered verdicts, agreeing with the human ranking on five of six system pairs (Kendall's $\tau = +0.67$).

A sub-ceiling per-item α would have been expected — but the assessor exceeded the ceiling. The literature reports LLM–human agreement frequently below human–human (Faggioli et al. 2023; Thomas et al. 2024), which is why the success criterion was set as the ceiling, not a fixed 0.7 bar, read against a mixed-expertise ceiling (two teachers, eight students). On the like-for-like anchor comparison the assessor agreed with individual teachers above it ($\alpha = 0.59$ vs. 0.46), so the domain-transfer validation succeeds for this setting. The honest counterweight is the dense/semantic tilt the assessor carries: part of that agreement may be shared semantic bias rather than shared correctness, which is why the assessor's qrels never enter the H1/H2 conclusions.

Circularity is contained, not eliminated. LLM relevance labels may tilt toward the dense/semantic channel, and reproducing a human ranking that already favours that channel is easier if the assessor shares the bias. This is why the ranking agreement is read together with the per-channel breakdown, and why the tilt is recorded as a limitation of the assessor. It never contaminates H1/H2, because those verdicts are computed on the human qrels and never on the LLM qrels — the single most important firewall in the evaluation design.

5.5 A worked failure case

One failure is documented concretely because it is visible directly in the labels and isolates a specific, fixable mechanism. Query `bac2025-e6` (a hard-edge, held-out BAC-2025 item) is a Diophantine problem — darts scoring 13 and 7 points, find the non-negative integer solutions of $13x + 7y = 1000$. Exactly one chunk was judged relevant: `bac-2024-maths-serie-c-0009`, the analogous worked solution from the 2024 paper, which is indexed (only 2025 is held out). So the relevant content is in the corpus.

The instructive point is which retriever found it. The dense channel retrieves the 2024 chunk at rank 1 (nDCG@5 = 0.83) — semantic similarity correctly matches the 2025 problem to its 2024 analogue despite different numbers. BM25 misses it (low lexical overlap between the two years' phrasings). And the RRF hybrid demotes it out of the top-5 (nDCG@5 = 0.00): with only one channel ranking the chunk highly, reciprocal-rank fusion averages dense's rank-1 against BM25's absent rank and the chunk falls below the cut-off. The lesson is a fusion failure, not a coverage gap — a single-query illustration of the §4.2 finding that the hybrid's advantage is carried by lexical overlap, and that on held-out items where query-to-corpus vocabulary diverges the dense channel is the more robust choice. It also shows the cost of RRF's channel-agnostic averaging — a chunk one strong channel ranks first can be buried when the other is silent — a concrete motivation for the score-aware or learned fusion flagged in future work.

5.6 Threats to validity

The threats consolidated in §M9 are restated here in order of how much they bound the conclusions.

Significance and robustness — now quantified. The paired query-level bootstrap is built: retrieval differences carry 95% CIs, and the H1 nDCG@5/nDCG@10 and H2 Recall@5 intervals exclude zero while the H1 Recall@5 and MRR intervals do not. Each verdict survives every label-perturbation scenario. With significance and label-robustness addressed, the largest remaining caveat on H1 is no longer statistical but one of external validity: the lexical-overlap finding means the hybrid advantage may be partly a property of the synthesised queries, and a real-student-query evaluation is required to settle whether it transfers.

Gold provenance and the mixed-expertise ceiling. The coverage qrels behind H1/H2 are single-rater and mostly student-produced, and the ceiling is mixed-expertise. The mitigation is the rubric: relevance is based on a correct human solution, model-independent, graded against fixed criteria applied uniformly — so trustworthiness rests on that anchor rather than the rater's role. A teacher↔student stratification is not pursued (the panel was not asked to adopt role-specific stances); the residual risk is disclosed, not removed.

Assessor circularity and paraphrase sensitivity. Contained by the human-qrels firewall (§5.4), with the dense-channel tilt and the single locked prompt's paraphrase sensitivity recorded as assessor limitations.

Contamination and leakage. BAC 2025 is de-indexed to limit leakage; the baseline smoke test ran on a small in-corpus subset and is calibration-only, not a result. Worked-example contamination is $\approx 32\%$ of the `theory`-labelled chunks (95% CI 20–48%, $n = 40$), about 12% of the theory bucket, which sizes the conservative bias on the H2-retrieval contrast; a full-corpus estimate is future work.

Generation faithfulness — the V2 silent-passthrough. The V2 strict-faithfulness prompt flags only errors the model has prior knowledge to detect; an error in content the model has not seen passes through silently. The temperature-0 prompt comparison (§M6) observed this directly: all three prompts reproduced a corrupted tile size without comment. On an OCR'd corpus where ~14.7% of chunks carry a hallucination flag, this is a real limitation of the generation component — documented, not a bug — and it bounds any future generation-quality claim, not the retrieval results here.

Corpus and scope limits. Figure-dropping (15.7% of chunks), the exclusion of geometry, and the single-textbook source bind every claim to non-geometry, text-only Terminale C content from one edition; a documented worked-example tagging gap bears specifically on H2. Roughly 100 queries and only four retrievers limit statistical power and generality.

5.7 What the work does and does not license

The work licenses three claims: that a curriculum-grounded retrieval-and-evaluation pipeline for Terminale C mathematics can be built and measured on real expert-adjacent labels; that the hybrid-vs-dense and pedagogical-vs-hybrid retrieval questions can be answered on this corpus;

and that the LLM-as-relevance-assessor method can be validated for this domain by ranking agreement and ceiling-relative per-item agreement.

It does not license any claim that the tutor generates better answers, that the theory–example mix improves teaching, that a frontier generator's grounding advantage over a small open model holds here (H4), or that students learn more. Each depends on the (query, response) gold or the human cohort that were out of reach for this submission. They are not gaps discovered late; they are the fully-specified next steps the foundation was built to enable, set out in Chapter 6.

Chapter 6. Conclusion and Future Work

6.1 Summary of contributions

This thesis set out to test four hypotheses about a curriculum-grounded retrieval-augmented tutor for Ivorian Terminale C mathematics. Under the constraints of a single-author MSc project and a fixed deadline, three were executed on data in hand and one (H4) was deferred; the executed core is a retrieval-and-evaluation foundation, reported honestly as such. The concrete contributions are:

1. A chunked, content-typed Terminale C mathematics corpus for a low-resource francophone curriculum — 722 chunks from the CIAM series with an OCR pipeline validated against a hand-transcribed sample, a content-type taxonomy, and a documented figure-dropping and leakage discipline (data card in Annex C).

2. A four-variant retrieval stack with two registered tests. BM25, dense (multilingual E5), their RRF hybrid, and a content-type-aware pedagogical retriever that enforces a theory-plus-worked-example mix. H1 was supported — hybrid beat dense by +0.049 nDCG@5 (95% bootstrap CI [+0.009, +0.093]), robust across every annotator-perturbation scenario, though the advantage is carried by the lexical channel and partly reflects overlap between the synthesised queries and the corpus. The H2 retrieval-side test was not met: the pedagogical retriever trailed hybrid slightly on relevance (−0.034 Recall@5, CI [−0.061, −0.007]) — a small, relevance-neutral cost of enforcing the theory–example mix, read as a necessary-condition check and not as the pedagogical-quality verdict, which per-chunk relevance cannot deliver.

3. A human relevance-annotation instrument and its agreement machinery — a 0–3 rubric based on a correct human solution, the disp10 overlap design over 1,565 (query, chunk) pairs labelled by a mixed panel of two teachers and eight Terminale C students, and tested Krippendorff's α / Cohen's κ implementations. The mixed-expertise inter-annotator ceiling is reported in §4.1.

4. A domain-transfer validation of the LLM-as-relevance-assessor method (the reframed H3): a protocol testing whether an LLM scoring (query, chunk) pairs on the same rubric can rank the four retrievers as the human qrels do (primary) and agree per-item near the human ceiling (secondary), with circularity contained by never letting the LLM qrels enter the H1/H2 verdicts. Verdict: the assessor (Claude Opus 4.8, 3,130 judgments over the 1,565 pairs) reproduced both human-qrels retrieval decisions — preserving H1 (hybrid >

dense) and H2-retrieval (pedagogical $\leq$ hybrid), agreeing with the human ranking on five of six system pairs (Kendall's $\tau = +0.67$) — and its per-item agreement with individual teachers ($\alpha = 0.59$ on the anchor) landed above the mixed-expertise human ceiling (0.46), with a documented dense/semantic tilt as its principal limitation.

A methodological contribution runs through all four: the BUILT-vs-PLANNED discipline and the H3 reframe (§5.1) — rebuilding an unexecutable hypothesis onto data in hand rather than reporting an under-powered or un-calibrated result — are themselves part of what the thesis demonstrates about rigorous evaluation under constraint.

6.2 Answering the research questions

RQ1/H1 (retrieval channel) — supported on this corpus. RRF fusion of sparse and dense retrieval beat dense-only by +0.049 nDCG@5, robust across every perturbation scenario; the qualification is that the gain is carried by the lexical channel and partly reflects vocabulary overlap between the synthesised queries and the corpus, so confirmation on real student queries is the natural next test. RQ2/H2 (pedagogical retrieval) — retrieval-side necessary condition met in the weak sense; the pedagogical-quality question is not answered here. The pedagogical retriever enforces its theory-plus-example mix at a small, bounded relevance cost (−0.034 Recall@5), which establishes only that the mix is achieved without large relevance loss — not that it tutors better, a verdict that per-chunk relevance cannot deliver and that awaits the generation study. RQ3/H3 (evaluation calibration) — supported. The calibrated LLM relevance assessor reproduced both human-qrels retrieval decisions (Kendall's $\tau = +0.67$) and agreed with individual teachers above the mixed-expertise human ceiling, with a dense/semantic tilt recorded as its principal limitation. RQ4/H4 (deployment feasibility) — deferred. The grounded frontier-versus-open generator comparison is fully specified but not executed, for want of the (query, response) gold and compute out of reach for this submission. The umbrella question — whether a curriculum-grounded RAG tutor for Ivorian Terminale C mathematics can be built and rigorously evaluated under real constraints — is answered affirmatively for its retrieval-and-evaluation foundation, with the generation-quality study its immediate continuation.

6.3 Future work

The future work is not a wish-list; it is the set of fully-specified next steps the foundation was built to enable, each blocked on a resource (human gold, a cohort, compute) rather than on an unanswered design question. They are listed in the order in which they most directly extend the executed core.

1. The response-grading judge (original H3 / H3b). The four-criterion judge — factual accuracy, curriculum alignment, pedagogical quality, language register (1–5), over complete generated responses — is already built and unit-tested but unrunnable for want of (query, response) gold, which requires a fresh round of human response-annotation. This is the single most immediate next step: it converts the relevance-assessor foundation into a response-quality evaluator. It is additionally being attempted as a delegated freeroll; nothing in this thesis depends on its outcome, but if it returns it upgrades to a reported H3b result.

2. The H2 generation side — the pedagogical-quality verdict. The central open question the retrieval-side H2 leaves unresolved (§5.3): does the enforced theory–example mix produce better tutoring answers, not merely a defensible retrieval set? This needs the response judge of (1) or the student study of (4), and it is the result that would let the thesis claim a teaching benefit rather than a retrieval property.

3. H4 — frontier vs. small open model under grounding. Whether grounding narrows the quality gap between a frontier generator (Claude) and a small open-weight model (e.g. Llama-3-8B, Gemma-2-9B) via a difference-in-differences over grounded vs. ungrounded responses, with latency and cost on a Pareto frontier. It depends on the response judge of (1) as its scoring instrument.

4. The student evaluation. A cohort of 10–20 Terminale C students, with a session protocol measuring correctness, pre/post confidence shift, and pairwise A/B preference between retrieval variants, plus qualitative interview coding. This is the only strand that measures learning rather than a proxy, and it requires consent (and parental consent for minors), a recruited cohort, and a piloted protocol — none compressible into the submission window.

5. Corpus and modality extensions. Lifting the scope limits: ingesting geometry (excluded here), handling figures rather than dropping them (a multimodal pipeline), broadening beyond the single textbook to test generalisation, and tuning the pedagogical retriever's (k_theory, k_example) split on a held-out slice with a reported Pareto frontier. A multi-turn tutoring setting is the natural system-level extension.

6. Evaluation-methodology hardening. Quantifying, at full-corpus scale, the contamination and worked-example tagging-gap estimates left pending (§5.6), beyond the n = 40 pilot bounds reported here.

6.4 Closing

The thesis does not claim a better tutor; it builds and validates the foundation on which that claim can be tested — a curriculum-grounded corpus, a measured retrieval stack, a human relevance instrument, and a validated LLM evaluator — for a setting (francophone, low-resource, curriculum mathematics) where such a foundation did not previously exist. The work it leaves undone is not a set of loose ends but a specified research programme, and the discipline by which an unexecutable hypothesis was reframed onto data in hand rather than abandoned or faked is, finally, the part most worth carrying forward.

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Table of Annexes

Annex	Title
A	Relevance rubric (v1.1) — the full 0–3 rubric, in French, as applied by the human panel and the LLM assessor
B	LLM relevance-assessor prompt — the locked system prompt, reproduced verbatim
C	Corpus data card — summary and pointer to `data/corpus/DATA_CARD.md`
D	Per-chapter corpus statistics
E	OCR validation — summary and pointer to the `marker` validation
F	Anonymised annotation sample — pointer to the returned relevance labels
M	Methodology, supplementary detail — decision rules and worked example (M.A), calibration pilot and rubric revision (M.B), overlap-design load balancing (M.C), generation- and student-evaluation specifications (M.D), retrieval implementation (M.E)

Annexes A–F are reproduced or summarised in the Annexes section; the methodology supplement (Annex M) carries the long-form design detail that the body of Chapter 3 points to. Page numbers for each annex are listed in the main table of contents above.

Annex M. Methodology — supplementary detail

This annex holds methodological detail moved out of Chapter 3 to keep the chapter to its page budget. Nothing here is new: each item is the long form of a design summarised in the body.

Annex M.A — Relevance rubric: decision rules and worked examples (expands §M7.3)

The 0–3 scale of §M7.3 is applied with seven decision rules that disambiguate the boundaries the calibration pilot showed were contested (§M7.5). In full:

1. Same-chapter floor. A same-chapter passage scores at least 1 even if a solution would not use it. 0 is reserved for a different chapter that contributes nothing.
2. Utility overrides topic upward. If a part of the passage genuinely helps with a part of the question it scores 2, even when it comes from another chapter, and even if a more direct passage exists elsewhere — a passage is not penalised because a better one was also retrieved.
3. Method, format, or procedure count (the stepping-stone rule). A passage exhibiting a step the question requires is useful (2) even if its own content is incomplete (e.g. an empty variation table for a function study), and central
   (3) if that method, applied, answers the whole question.
4. Coverage separates 2 from 3. Helping one part scores 2; sufficing for all parts scores 3.
5. Each passage is judged alone, never by comparison with the others retrieved for the same query.
6. Conceptual questions. For an "explain why…" question, a passage giving the explanation, intuition, or proof scores 3, whereas a bare calculation or incidental property scores 1.
7. Worked examples accompany the rubric and are reproduced (with the rubric itself, in French) in Annex A; the four-level worked example is drawn from the integral-parity pilot cluster (§M7.5).

Annex M.B — Calibration pilot and the v0.1 → v1.0 → v1.1 revision (expands §M7.5–§M7.6)

The pilot. A calibration pilot (master task 15) had two teachers independently grade 80 pairs drawn from 5 stratified queries under rubric v0.1. The pilot was diagnostic, not a result: its labels are calibration data, explicitly not the final gold.

What it showed. Weighted κ on the graded scale was 0.704, with 96.2% adjacent agreement ($|\text{difference}| \leq 1$) and only three pairs off by two — no wild disagreement. The binary-relevant (≥ 2) κ was 0.587, the weak point, and it mattered because ≥ 2 is the Recall threshold. Of 32 disagreements, the 1↔2 boundary dominated (13), ahead of 0↔1 (8) and 2↔3 (8), clustering on property-heavy, multi-step queries (an integral-parity BAC item and a function-study item accounted for roughly half). The diagnosis: v0.1 did not tell teachers how to score a passage that is "right chapter, similar-looking, but not the property the question needs."

The v1.0 revision (locked 2026-06-18 for the full dispatch) responds point-for-point:

1. reframes the guiding question to "would a correct solution use this?" and adds the "je m'en sers" test for the 1↔2 boundary — the principal fix;
2. sharpens 0↔1, reserving 1 for same-lesson-but-unused and sending different-chapter-contributes-nothing to 0;
3. sharpens 2↔3, with 3 = the exact sufficient property/method and 2 = a necessary piece needing others;
4. adds the stepping-stone rule (rule 3, Annex M.A);
5. adds the real four-level worked example drawn from the integral-parity pilot cluster.

The pilot evidence and each change are recorded in the rubric-v1.0 decision record.

The v1.1 written form (2026-06-27). At dispatch the written v1.0 was accompanied by oral clarifications on the two contested boundaries: a same-chapter passage that goes unused scores 1, not 0, and partial utility — a part of a passage serving a part of the question, including a required method or format — scores 2, with full coverage scoring 3. Version v1.1 writes these clarifications down. It does not alter the underlying definition (utility for a correct solution); it fixes its application on those two boundaries, and it is the version the LLM relevance-assessor is prompted with (§M8), so machine and panel judge against the same applied definition. The human gold reflects v1.0 as briefed — the same clarifications were given to all annotators at dispatch; v1.1 is its faithful written form, not a post-hoc change of standard.

Annex M.C — Overlap design: load balancing, the design revision, and the disp05 contingency (expands §M7.9–§M7.10)

The binding constraint is per-annotator load, not headcount. With the queries to be covered once and a small anchor budget, the design question is how many quick 0–3 judgements each annotator must make. The design was revised once on exactly this axis. The first cut — a

20-query × 5-rater anchor plus 80 singly-graded coverage queries — placed about 283 pairs on each annotator, judged too heavy. It was rebalanced to the 10-query × 2-rater anchor plus 90 coverage queries used for dispatch, cutting per-annotator load by roughly 39% to about 11 queries / ~172 pairs each. Every query is still covered exactly once (covering all 100 is why each annotator carries ~11 queries, not 10). The two trade-offs this revision accepted are explicit: anchor redundancy dropped from 5 to 2 raters, and the anchor's chapter span narrowed from 10 to 7 (§M7.8) — both justified by the pilot's $\alpha = 0.70$ at redundancy 2.

Allocation mechanics. Within a tier, load is balanced to within ±7% by greedy pair-count allocation, and the two anchor raters per query are assigned by a sliding window so that which pair of annotators co-rates varies across the anchor — giving the agreement estimate broader coverage of rater pairings than a single fixed pair would. The whole assignment is generated deterministically (seed 42), so any design variant is exactly reproducible by re-running the generator.

The realised dispatch (disp10) and the contingency (disp05).

Option	Annotators	Anchor	Coverage	Per-annotator	Total judgements
disp10 (realised)	10	10 × 2 raters	90 × 1	~11 queries, 165–179 pairs	1,720
disp05 (contingency, not used)	5	10 × 3 raters	90 × 1	~24 queries, 369–380 pairs	1,877

The fifteen dispatch identifiers (T01–T15) recorded in the 2026-06-24 handoff do not indicate fifteen independent grading loads: ten annotators completed the dispatch, and five further annotators who became available later re-filled five of the ten assignment files already present (re-grading those slots rather than extending coverage). Coverage and the design are therefore those of the ten-person disp10; the five refills are replacement labour on five existing assignments, not a sixth-through- fifteenth tier.

Annex M.D — Generation-side and student evaluation: full specifications (expands §M10)

Both evaluations below are designs, not results. Both depend on a (query, response) gold set that was never collected (only the (query, chunk) relevance package was dispatched; §M7.1, §M8), so neither could be run before submission. They are the immediate next steps the executed retrieval-and-evaluation foundation was built to enable, and are carried as future work in Chapter 6.

Annex M.D.1 — Generation-side evaluation (H2 generation side; the response judge, H3b)

The instrument is the response judge — the original H3, scoring a full tutor response on four model-independent criteria (factual accuracy, curriculum alignment, pedagogical quality, language register), 1–5 each, variant-blind, based on a correct human solution to avoid generator–judge circularity (§2.4, C2). Its scoring harness is built and unit-tested (Opus 4.8, schema-constrained), but is unrunnable: it needs teacher response-gold to calibrate against, and the four-criterion rubric descriptors are still draft v0.1 (task #13). The specified protocol:

1. finalise the four-criterion rubric;
2. generate ~150 (query, response) pairs and dispatch response-grading annotation;
3. calibrate the judge against that gold (judge–teacher Krippendorff's α / per-criterion κ, plus judge test–retest reliability);
4. score responses generated by all four retrieval variants and compare (paired tests, §M9).

This is what would test the H2 generation-side claim — does the theory-plus-example mix yield better answers, not just better retrieval — which §M8 is explicit cannot be concluded from the retrieval metrics alone. Grounded-vs-ungrounded scoring under this same judge is also the instrument for H4. Status: harness built; calibration and run not executed.

Annex M.D.2 — Student evaluation (field validation)

A small field study with 10–20 Terminale C students, validating the tutor with the target population rather than against labels. Specified design:

- Session: a fixed problem set, scored for correctness, with pre/post confidence ratings and a short interview; a pairwise A/B in which the student compares pedagogical vs. standard-hybrid answers blind — structurally a small preference arena (cf. §2.4) that yields the student-side H2 signal.
- Run: consent (with parental consent for minors); a 1–2 student pilot to validate the protocol; then full sessions.
- Analysis: quantitative (correctness, confidence shift, A/B preference rate) and a light qualitative coding of the interviews.

No part is built, and the study needs a human cohort and ethics turnaround it could not get before the deadline. Status: specified, not executed.

Annex M.E — Retrieval implementation detail (expands §M4)

The three retrieval channels of §M4 are run at untuned defaults; their implementation specifics are recorded here.

BM25 (sparse). Okapi BM25 via `rank-bm25` at library defaults ($k_1 = 1.5$, $b = 0.75$) — fixed, like the slot sizes of §M5.2, so the H1 contrast reflects the value of adding a sparse channel under standard conditions rather than of a tuned one. Tokenisation uses the spaCy French pipeline

[Honnibal & Montani, 2017] with three maths-motivated adjustments: (1) single-character variables preserved — the stopword filter would otherwise drop length-1 tokens such as "a", "x", "n", "k", which are content-bearing and often the strongest lexical signal in maths; (2) accent folding (é→e, à→a, ç→c), so an accent-less query ("theoreme") still matches accented corpus text — a lexical-channel choice only, since the dense channel resolves diacritics in its own subword tokeniser; (3) no lemmatisation, which would wrongly conflate "limites" (noun) with "limité" (participle). The index is built once and not rebuilt across experiments.

Dense (multilingual E5). `intfloat/multilingual-e5-large`, a 1024-dim multilingual encoder. Raw passage and query text is fed to its pretrained subword tokeniser, which is why the BM25 pre-processing applies to the sparse channel only. Per the E5 recipe, queries carry prefix `query:` and passages `passage:` (deviating measurably degrades quality); both embeddings are L2-normalised and compared by cosine, computed once at index time. The encoder is not fine-tuned on CIAM: the 100-query set is too small to fine-tune without overfit, and H1 concerns retrieval channels, not encoder adaptation — fine-tuning would confound the channel-level comparison.

RRF hybrid. The hybrid fuses the BM25 and dense rankings by Reciprocal Rank Fusion [Cormack et al., 2009]: $\mathrm{rrf}(d) = \sum_{r \in \{\mathrm{BM25,dense}\}} 1/(60 + \mathrm{rank}_r(d))$, with canonical $k = 60$. Each retriever contributes its top-20; the union, re-ranked by the fused score, is the RRF-fused pool, consumed both by the hybrid (top-5 in fused order) and by the pedagogical retriever (the same pool, a different selection policy; §M5.3). RRF is preferred over learned fusion because the labelled data is insufficient to train a fuser without overfit, so the H1 contrast reads as dense-only vs a standard, untuned hybrid. # Annexes

Supporting artefacts referenced from the body. Each annex is either reproduced here in full (where the thesis must be self-contained — the relevance rubric and the locked assessor prompt) or is a pointer to the canonical, version-controlled artefact in the repository (the data card, the OCR validation, the per-chapter statistics), with a short summary so the annex stands on its own. The methodology annex (long-form design detail) is Annex M.

The annexes are:

- Annex A — Relevance rubric (v1.1): the 0–3 relevance rubric, in French, exactly as applied by the human panel and the LLM assessor.
- Annex B — LLM relevance-assessor prompt: the locked system prompt given to the assessor (§M8, §4.5). It embeds Annex A verbatim, so the two are presented together.
- Annex C — Corpus data card: pointer + summary of the corpus data card.
- Annex D — Per-chapter corpus statistics: per-document composition table.
- Annex E — OCR validation : pointer + summary of the marker validation.
- Annex F — Anonymised annotation sample : pointer to the returned labels.

Annex A & B — Relevance rubric (v1.1) and the LLM assessor prompt

The human panel (§M7) and the LLM relevance-assessor (§M8, §4.5) judged (query, chunk) relevance against the same rubric — that identity is the methodological point of the reframed H3, so the rubric and the assessor prompt are one artefact. What follows is the locked assessor system prompt verbatim: an instruction line, the full v1.1 rubric (the same document the panel used, with the dispatch's verbal clarifications written down), and the grading task. Its sha256 (`prompt_id = 81a7d8a86e0c2008`) is recorded in every assessor run, so the exact text below is provably the one that produced the §4.5 results.

The rubric is in French (the data, prompts, and rubric language; CLAUDE.md): the panel and the model both work in the language of the curriculum.

```
Tu es enseignant(e) de mathématiques de Terminale C (Côte d'Ivoire) et tu annotes la per

# Rubrique de pertinence - annotation (version clarifiée)

**Version : v1.1** (formalisation écrite des consignes verbales de la dispatch - voir §

Pour chaque paire **(question d'élève, passage du cours retrouvé)**, on attribue une **n

## Question à se poser

> « **Pour répondre correctement à cette question, une solution rédigée utiliserait-elle

On juge **ce passage précis**, par rapport à **cette question précise** - pas le chapitr

> **Test « je m'en sers »** (repère pratique) : imaginez que vous rédigez la solution de

## Granularité du « thème »

Le **thème** se juge au **niveau du chapitre** du programme (p. ex. « fonctions exponent

## Échelle de notation (0-3)

| Note | Sens |
|---|---|
| **3 - Suffisant** | Les informations du passage **suffisent à elles seules** à répondr
| **2 - Utile** | **Une partie** du passage **aide clairement** à résoudre **une partie*
| **1 - Marginal** | **Même chapitre / même thème** que la question, mais **aucune parti
| **0 - Non pertinent** | **Autre chapitre / autre thème ET rien** dans le passage n'aid

## Règles de décision

1. **Le « bon thème » garantit au moins 1, jamais 0.** Un passage du même chapitre que l
```

```
2. **L'utilité prime sur le thème (vers le haut).** Si **une partie** du passage aide ré
3. **Méthode / format / procédé comptent.** Un passage qui montre **la méthode ou le for
4. **Même méthode, autres nombres → jusqu'à 3.** Un exemple traité qui montre directemer
5. **Couverture pour distinguer 2 et 3.** **2** = aide pour **une partie** de la questio
6. **On juge le passage seul**, jamais par comparaison avec les autres passages retrouvé
7. **Question conceptuelle** (« explique pourquoi… ») : un passage qui donne l'explicati

## Exemples courts

Question : « *Résous dans ℝ : e^(2x) − 5·e^x + 6 = 0* »

| Passage | Note | Pourquoi |
|---|---:|---|
| Exemple traité résolvant `e^(2x) + e^x − 2 = 0` en posant X = e^x | **3** | même métho
| Propriété « e^x > 0 pour tout réel x » (sert à écarter une racine) | **2** | une parti
| Limites de l'exponentielle (même chapitre, non utilisé) | **1** | bon thème, mais aucu
| Passage de probabilités | **0** | autre chapitre, aucun apport (définition de 0) |

Question : « *Fais l'étude de la fonction f(x) = e^(x² + ln x)* »

| Passage | Note | Pourquoi |
|---|---:|---|
| Tableau de variation **vide** (gabarit) | **2** | le tableau de variation **fait parti
| Méthode complète d'étude de fonction (domaine, dérivée, variations, limites) | **3** |
| Calcul d'une probabilité conditionnelle | **0** | autre chapitre, aucun apport |

## Seuil pour les métriques (rappel technique, pas pour l'annotateur)

- **Recall@k** : un passage est compté « pertinent » si **note ≥ 2** (une partie au moi
- **nDCG@k** : utilise les notes graduées 0-3 telles quelles.

## Correspondance avec l'échelle standard (validité)

Cette échelle 0-3 reprend, en l'adaptant au domaine, l'échelle graduée de pertinence des

## Versionnement

- Chaque annotation est **horodatée avec la version de rubrique** en vigueur (champ `rub
- **v1.0** (2026-06-18) est la version verrouillée du **document écrit** remis pour la c
- **v1.1** (2026-06-27) **écrit ces consignes verbales** ; elle ne change pas la définit
- **Ne pas modifier la rubrique après coup** sans incrémenter la version.

## Confidentialité

Les noms des annotateurs n'entrent jamais dans le dépôt : utiliser des identifiants anon
```

```
Pour la paire (question, passage) fournie, attribue UNE note de pertinence de 0 à 3 en a
```

Annex C — Corpus data card (summary + pointer)

The full data card (v2.2, snapshot 2026-06-10, BAC-2025 held out 2026-06-17) is written to the spirit of Gebru et al. (2021) and Mitchell et al. (2019). The headline facts:

- Composition. 722 chunks from 12 indexed documents — ten CIAM Terminale C chapters (706 chunks) plus two BAC série C papers (2022, 2024; 16 chunks). BAC 2025 is held out, not indexed (leakage prevention; §M3.5). 424,010 prose characters, 75,853 words, 5,313 equations.
- Content-type taxonomy (8 values). theory 185 (25.6%), theorem 142 (19.7%), example 141 (19.5%), remark 82 (11.4%), proof 57 (7.9%), definition 48 (6.6%), exercise 34 (4.7%), solution 33 (4.6%).
- Provenance. Source PDFs (CIAM, EDICEF/NEI; BAC papers, MENA Côte d'Ivoire) are gitignored (copyright); the corpus is research-only. OCR by `marker`, with degeneration-loop cleaning and section-aware chunking.
- Known limitations (stated loudly): figures dropped (15.7% of chunks); geometry excluded (figure-dependent); equation error tail (~12.5% below 0.95 similarity); `theory` is a fallback bucket carrying headingless worked examples (see Annex D note and §4.3/§5.3); OCR validation is method-transfer from one hand-checked chapter, not per-document.

Annex D — Per-chapter corpus statistics

Per-document composition (source: the corpus data card, §2.2).

Document	Topic	Chunks	Eq.	Fig-drop	Flagged
`6_ciam_tc`	Arithmétique dans Z	137	511	8	11
`56_ciam_tc`	Nombres complexes	87	732	17	13
`172_ciam_tc`	Dénombrement et probabilités	70	323	6	5
`194_ciam_tc`	Limites et continuité	67	568	19	8
`214_ciam_tc`	Dérivation	53	446	7	14
`234_ciam_tc`	Primitives et logarithme népérien	67	580	9	9
`256_ciam_tc`	Fonction exponentielle	70	482	12	18
`276_ciam_tc`	Suites numériques	51	590	6	7
`296_ciam_tc`	Calcul intégral	71	487	24	14
`322_ciam_tc`	Équations différentielles	33	451	5	5
`bac-2022-maths-tc`	BAC série C 2022	7	89	0	1
`bac-2024-maths-serie-c`	BAC série C 2024	9	54	0	1
held out: `bac-2025-maths-tc`	BAC série C 2025	6	54	0	—

Annex E — OCR validation (summary + pointer)

OCR quality was validated by hand-transcribing one CIAM chapter (Ch. 2, divisibility) and scoring `marker` against it: prose CER 11.51%, prose WER 14.40%, equation-detection F1 87.13%, equation exact-match 87.46%. A second engine (`docling`) underperformed `marker` by an order of magnitude on equations. v2 applies the validated method to the full document set without re-collecting ground truth (a documented method-transfer assumption; data card §5.7).

Annex F — Anonymised annotation sample (pointer)

The returned human relevance labels (disp10 design; 1,720 labels over 1,565 pairs; §M7.10) are identified only by anonymised codes `T01`–`T10` — no names enter the repository. The LLM assessor's labels are recorded under `annotator_id` ∈ {`LLM_run1`, `LLM_run2`}.